Understanding context in menthol cigarette bans: A systematic evidence synthesis using causal loop diagramming



UNC GILLINGS SCHOOL OF GLOBAL PUBLIC HEALTH

UNC LINEBERGER COMPREHENSIVE CANCER CENTER

Clayton Ulm, BS; Sarah D. Mills, PhD, MPH; Maya Wilson, MS; Frantasia Hill, MPH; Evan Liebgott, MPH; Anne Donnelly, MPH; Shelley D. Golden, PhD, MPH; Kurt M. Ribisl, PhD; Kristen Hassmiller Lich, PhD

Disclosures and funding

Advancing Tobacco Regulatory Science to Reduce Health Disparities, UNC TCORS (NIH/FDA U54DA060049)

Disclosures

KMR has served as a paid expert consultant representing plaintiffs in litigation against e-cigarette and tobacco companies. All other authors have no conflicts of interest.

Background

- Analyses of menthol cigarette bans often assume bans are “on” or “off”

Background

- Analyses of menthol cigarette bans often assume bans are “on” or “off”
- Contextual factors influence policy outcomes
    - *Enforcement delays and tobacco industry tactics to circumvent bans*

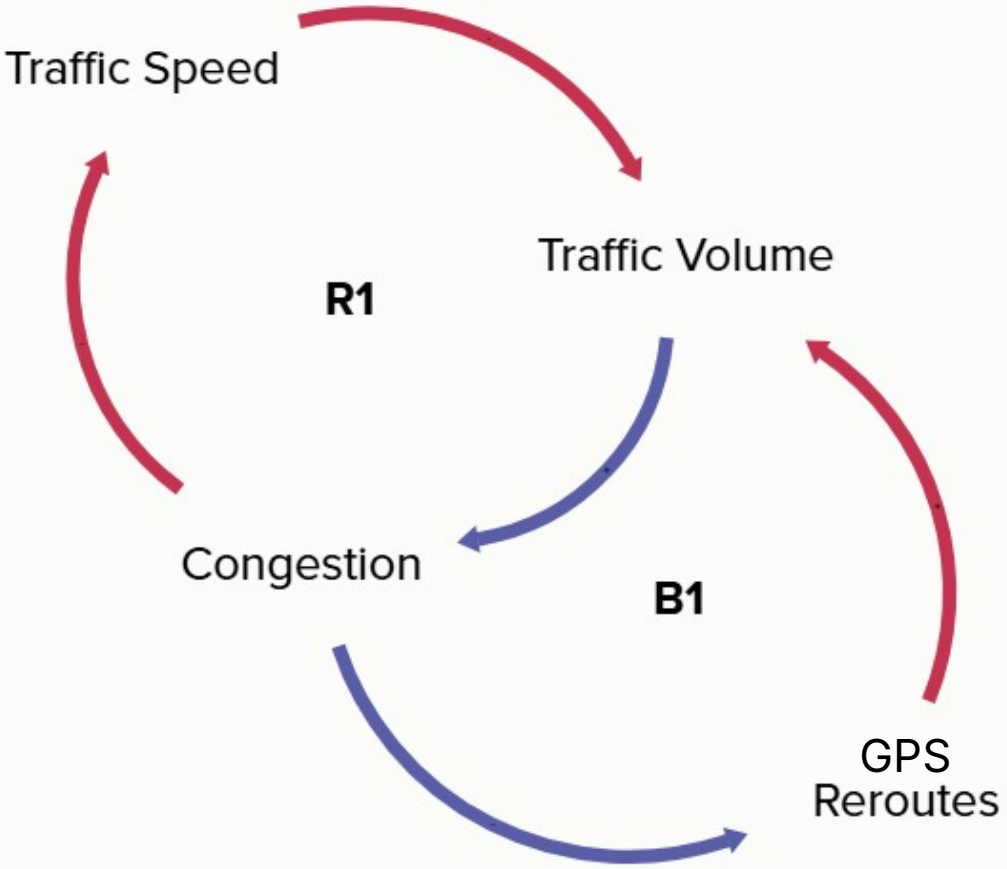
Objective

- Understand key contextual factors that may impact the effectiveness of menthol cigarette bans

Methods

- We developed causal loop diagrams (CLDs) to visualize system dynamics of a menthol ban
- CLDs consist of variables and arrows, which connect to form loops that are reinforcing or balancing

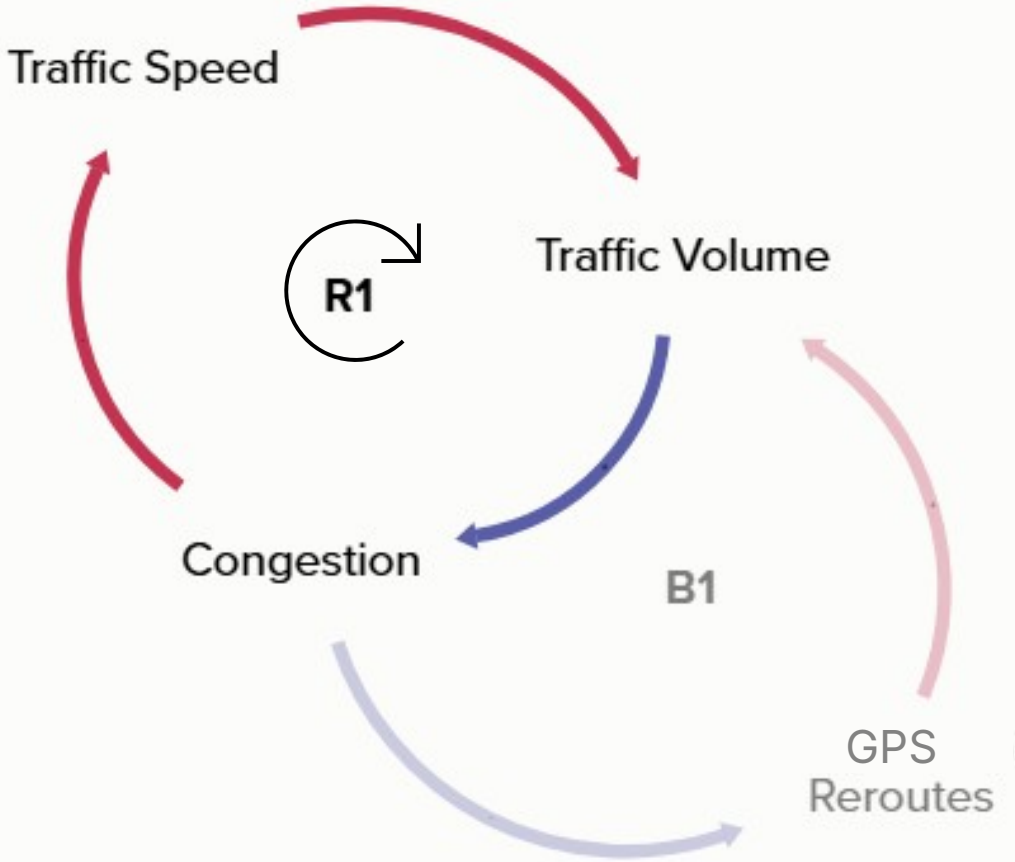
Causal Loop Diagramming



Same

Opposite

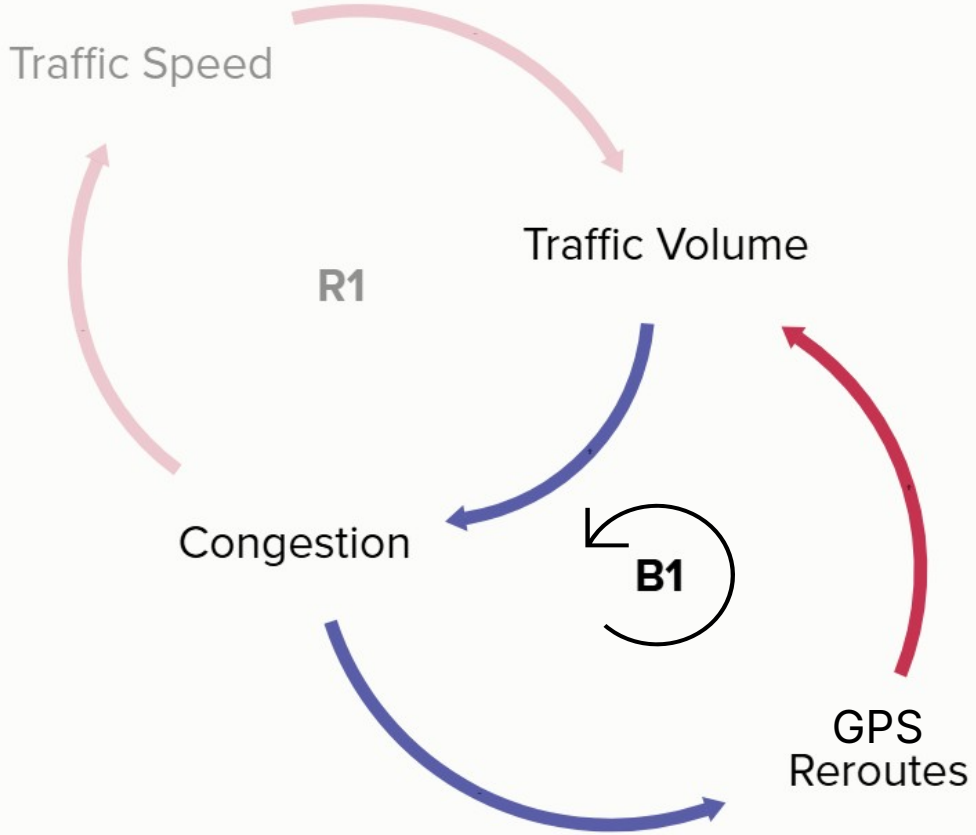
Causal Loop Diagramming



Same

Opposite

Causal Loop Diagramming



Same

Opposite

Process

Identify literature related to menthol bans

Extract causal linkages and code variables

Visualize and synthesize causal linkages

Identify literature related to Menthol Bans

Systematic Reviews

Flavored Cigar Ban

***K* = 55**

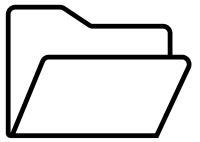
+

Menthol Cigarette Ban

***K* = 110**

→

141 Articles



Process

Identify literature related to menthol bans

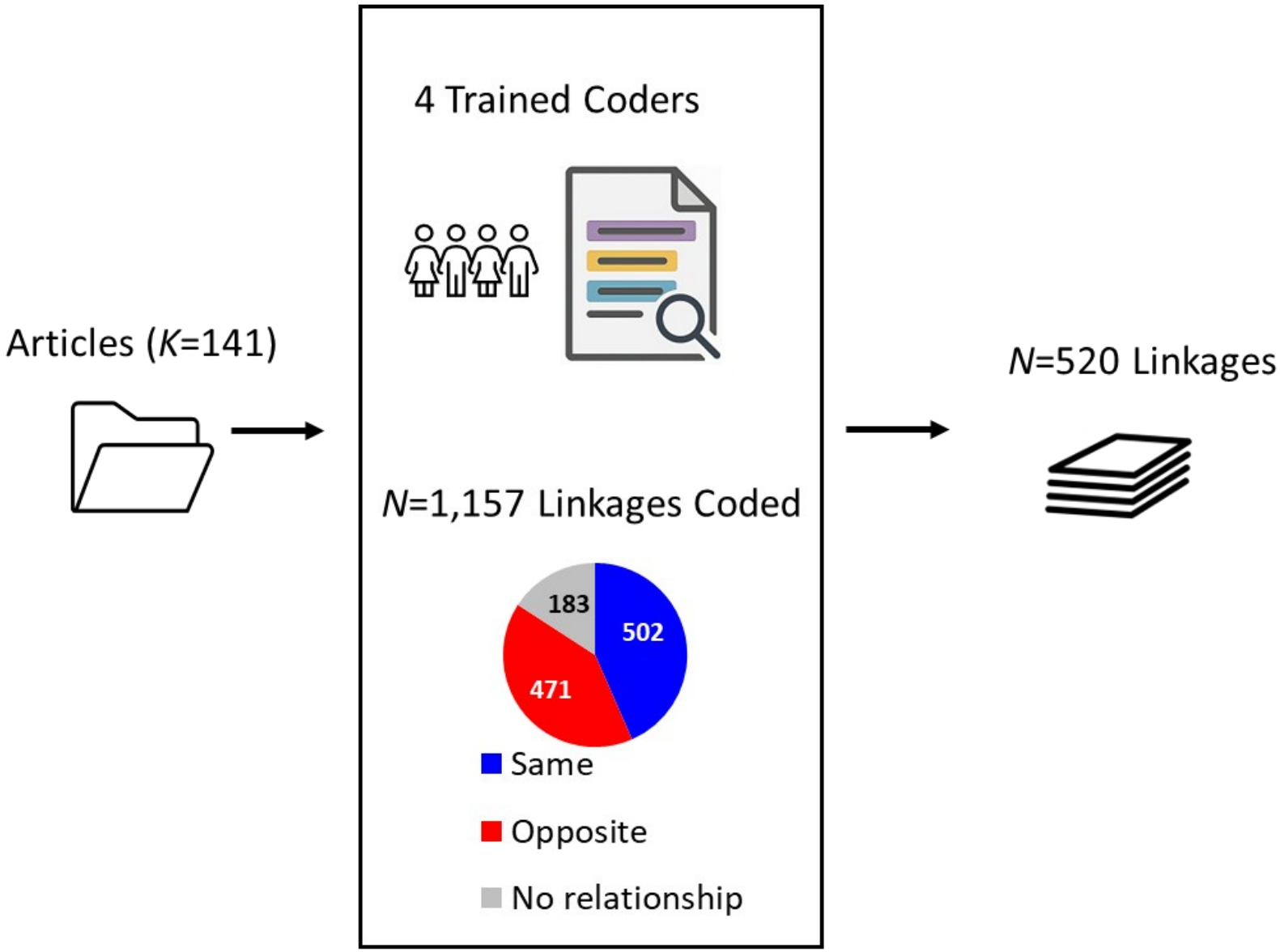
Extract causal linkages and code variables

Visualize and synthesize causal linkages

Extract causal linkages from the literature

- In each article, we coded:
  - Cause → Effect Statements
  - Polarity: **Same** (+), **Opposite** (–), **Independent** (=)

Causal Linkages Extraction



Process

Identify literature related to menthol bans

Extract causal linkages and code variables

Visualize and synthesize causal linkages

Key Narratives



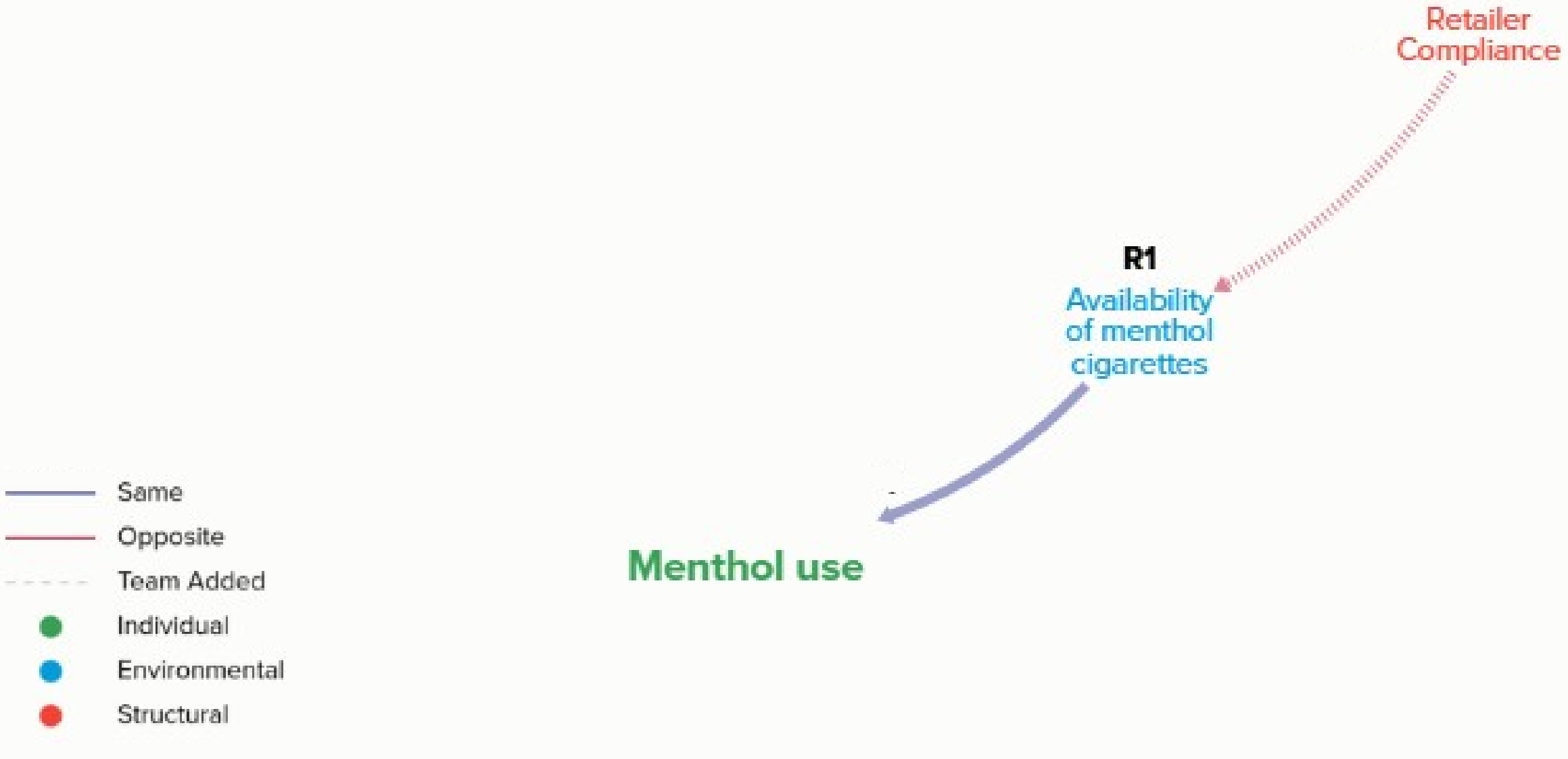
Retailer Compliance

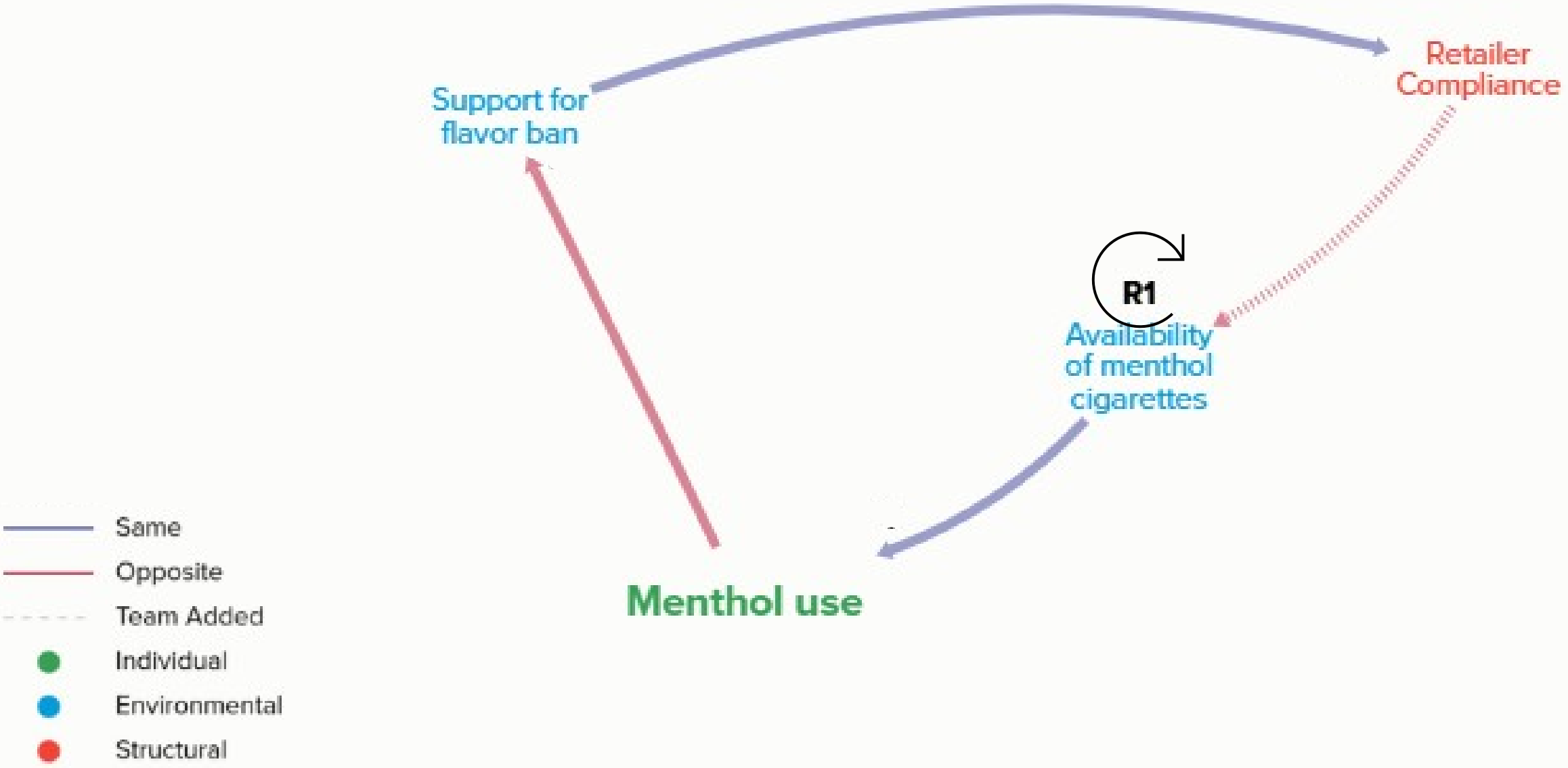


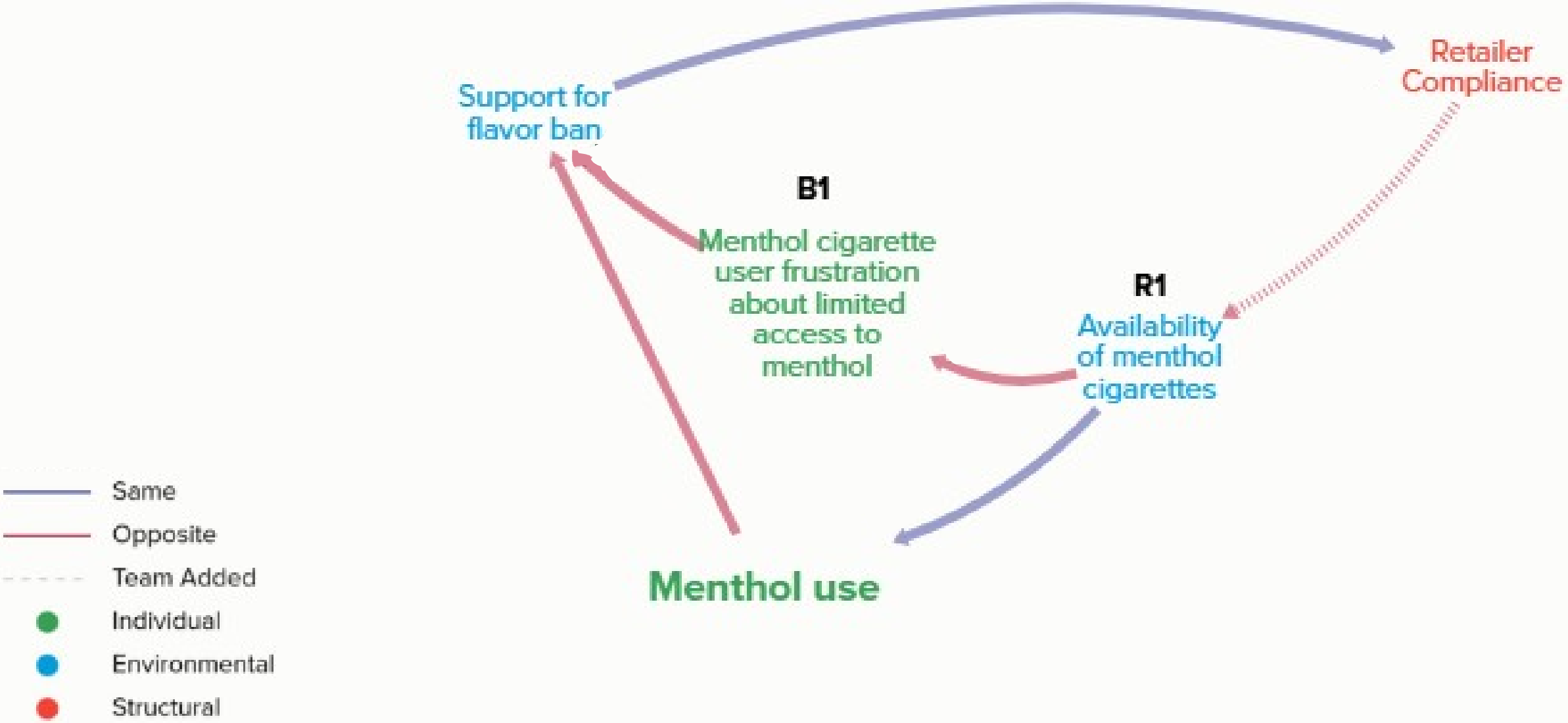
Industry Tactics

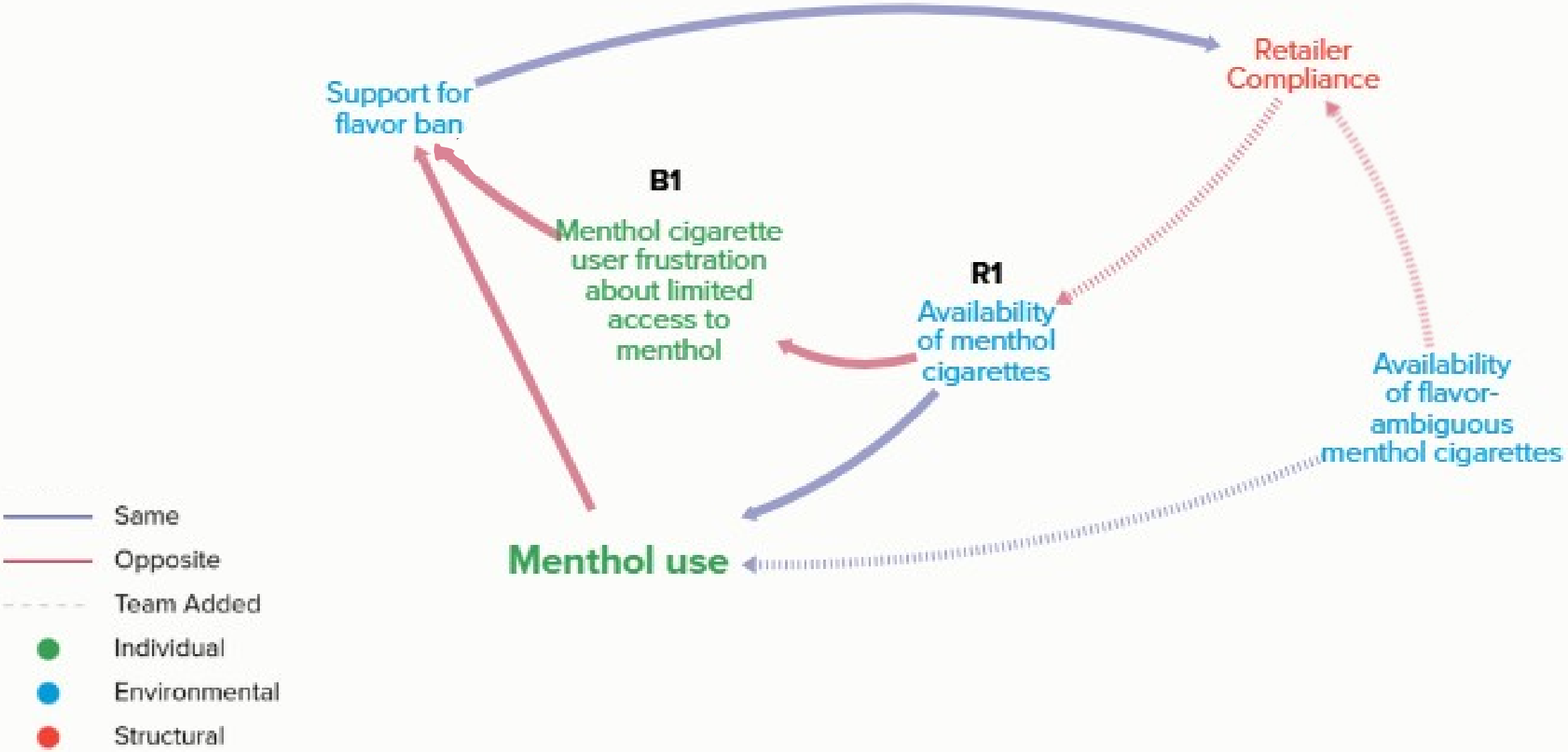


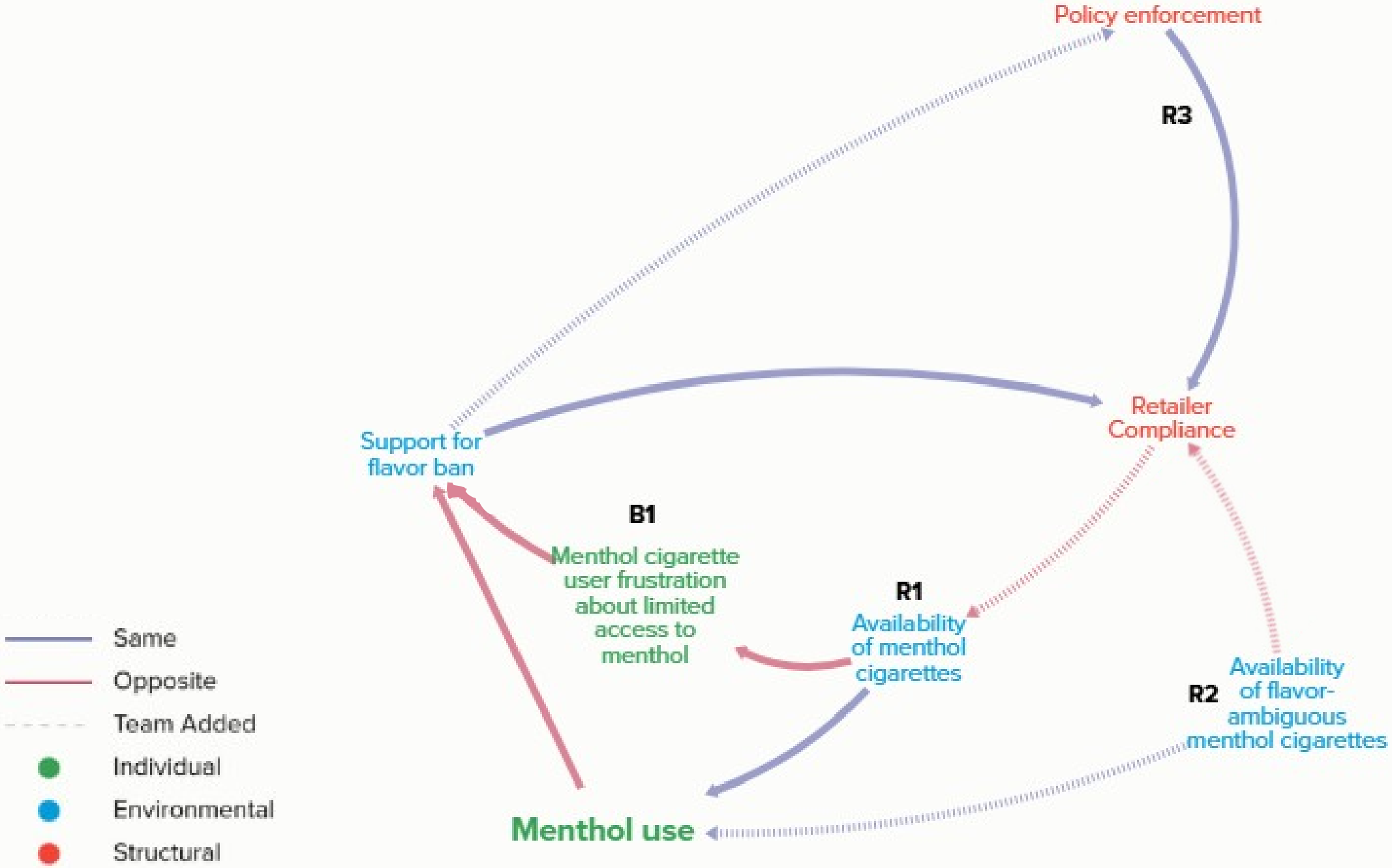
Impacts of a Ban

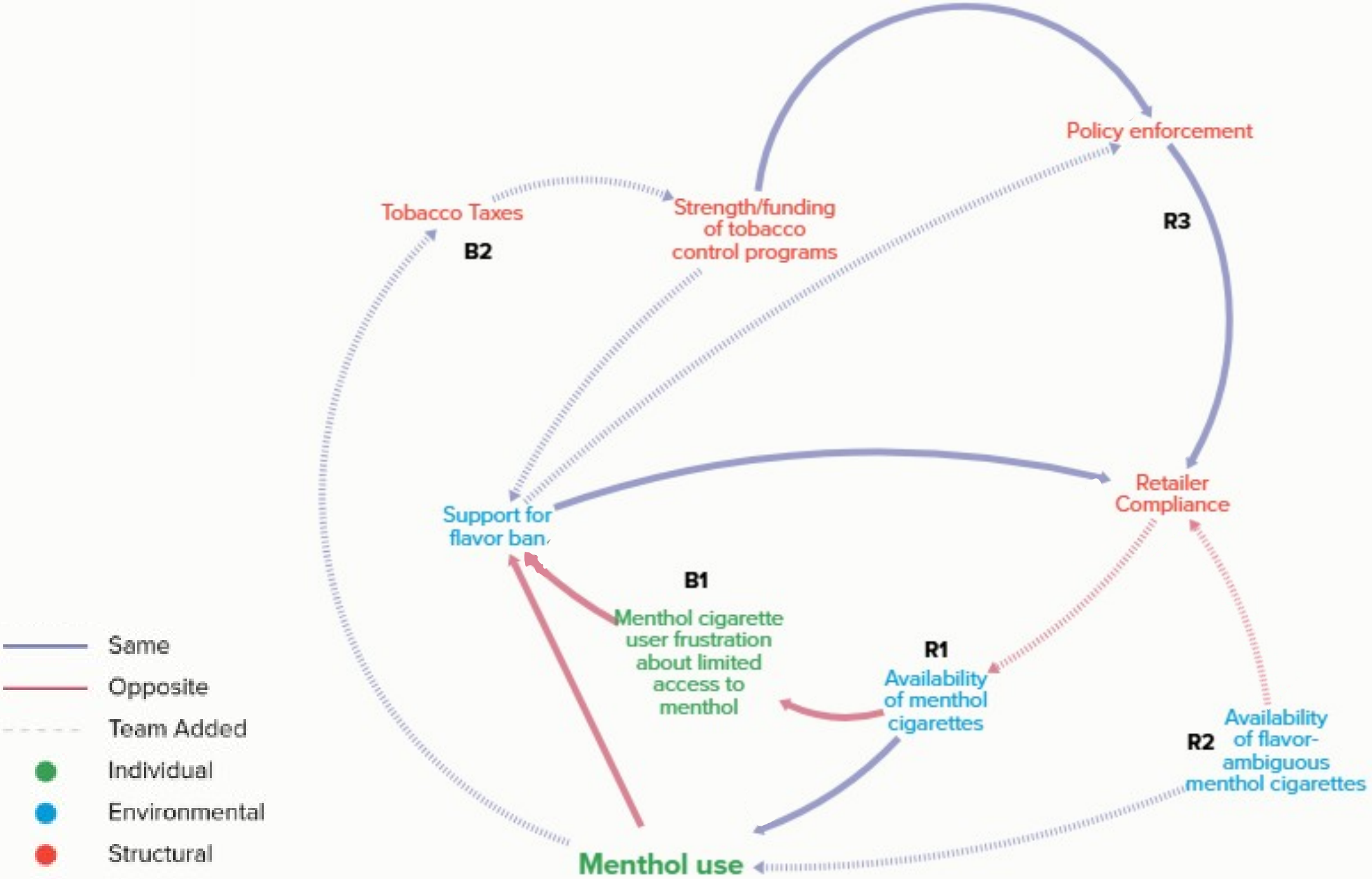

Retailer Compliance
R1
Availability of menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Support for flavor ban
Retailer Compliance
R1
Availability of menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Support for flavor ban
Retailer Compliance
B1
Menthol cigarette user frustration about limited access to menthol
R1
Availability of menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Support for flavor ban
Retailer Compliance
B1
Menthol cigarette user frustration about limited access to menthol
R1
Availability of menthol cigarettes
Availability of flavor-ambiguous menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Policy enforcement
R3
Retailer Compliance
Support for flavor ban
B1
Menthol cigarette user frustration about limited access to menthol
R1
Availability of menthol cigarettes
R2
Availability of flavor-ambiguous menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Policy enforcement
Tobacco Taxes
B2
Strength/funding of tobacco control programs
R3
Retailer Compliance
Support for flavor ban
B1
Menthol cigarette user frustration about limited access to menthol
R1
Availability of menthol cigarettes
R2
Availability of flavor-ambiguous menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural

Narrative Conclusions

- Strong public support and enforcement enhances compliance and lowers menthol availability

- User frustration reduces support and lowered taxes can balance this effect over time

Key Narratives



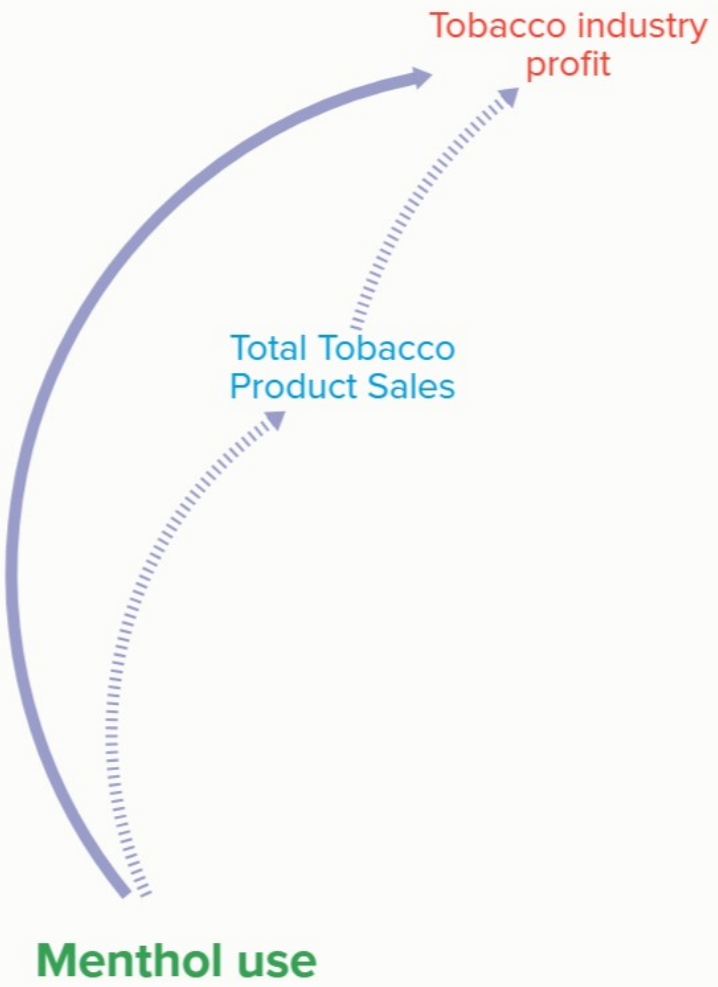
Retailer Compliance



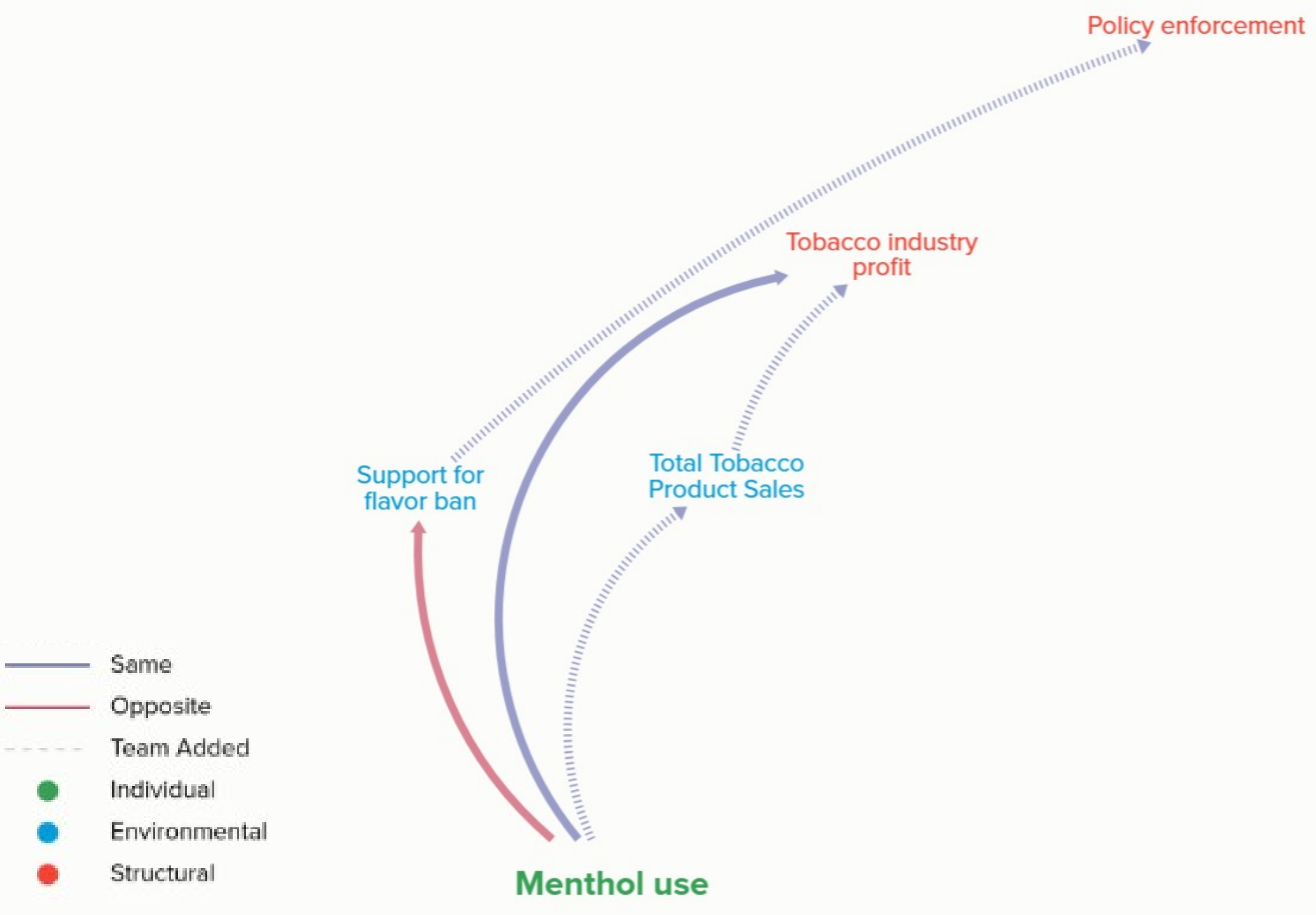
Industry Tactics

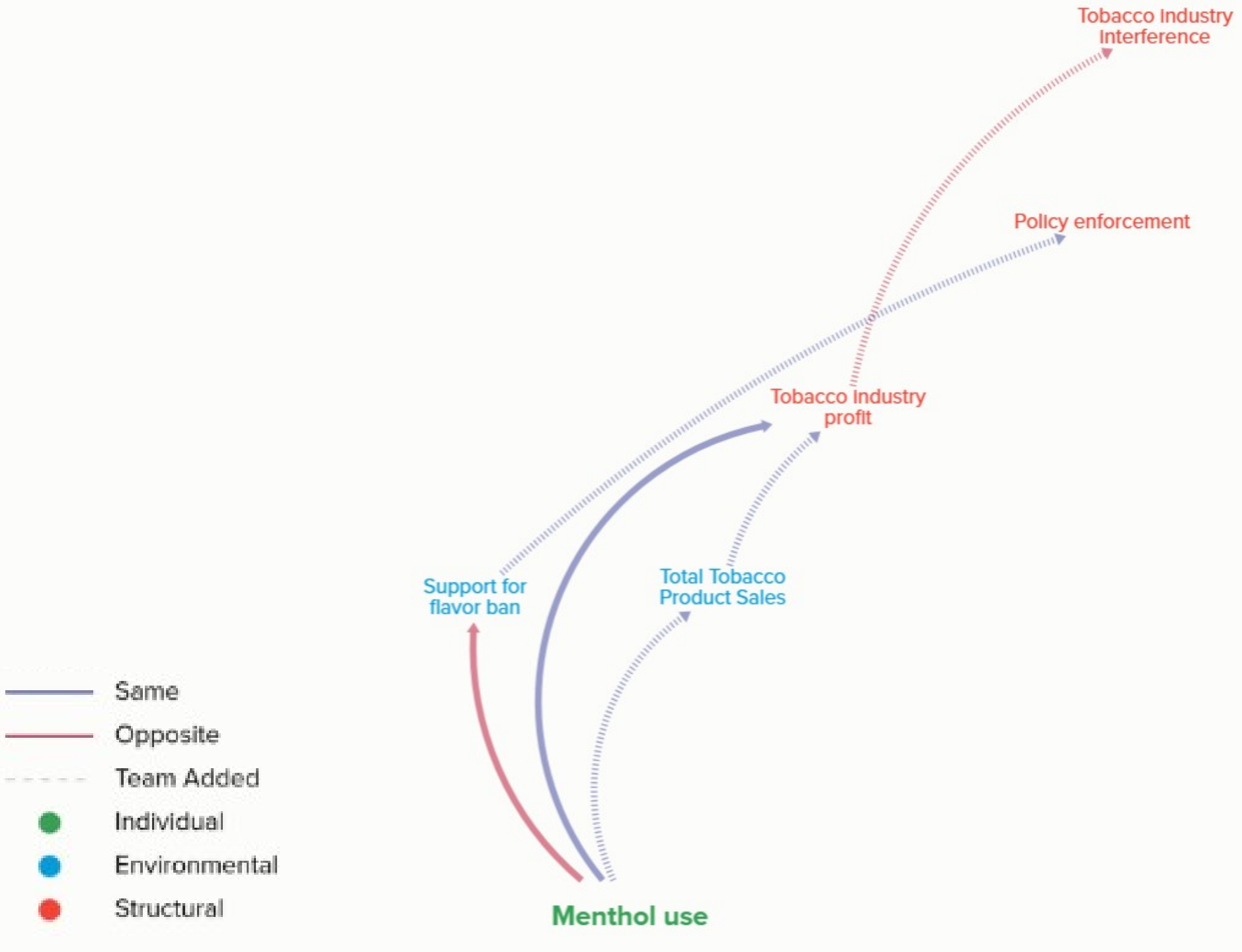


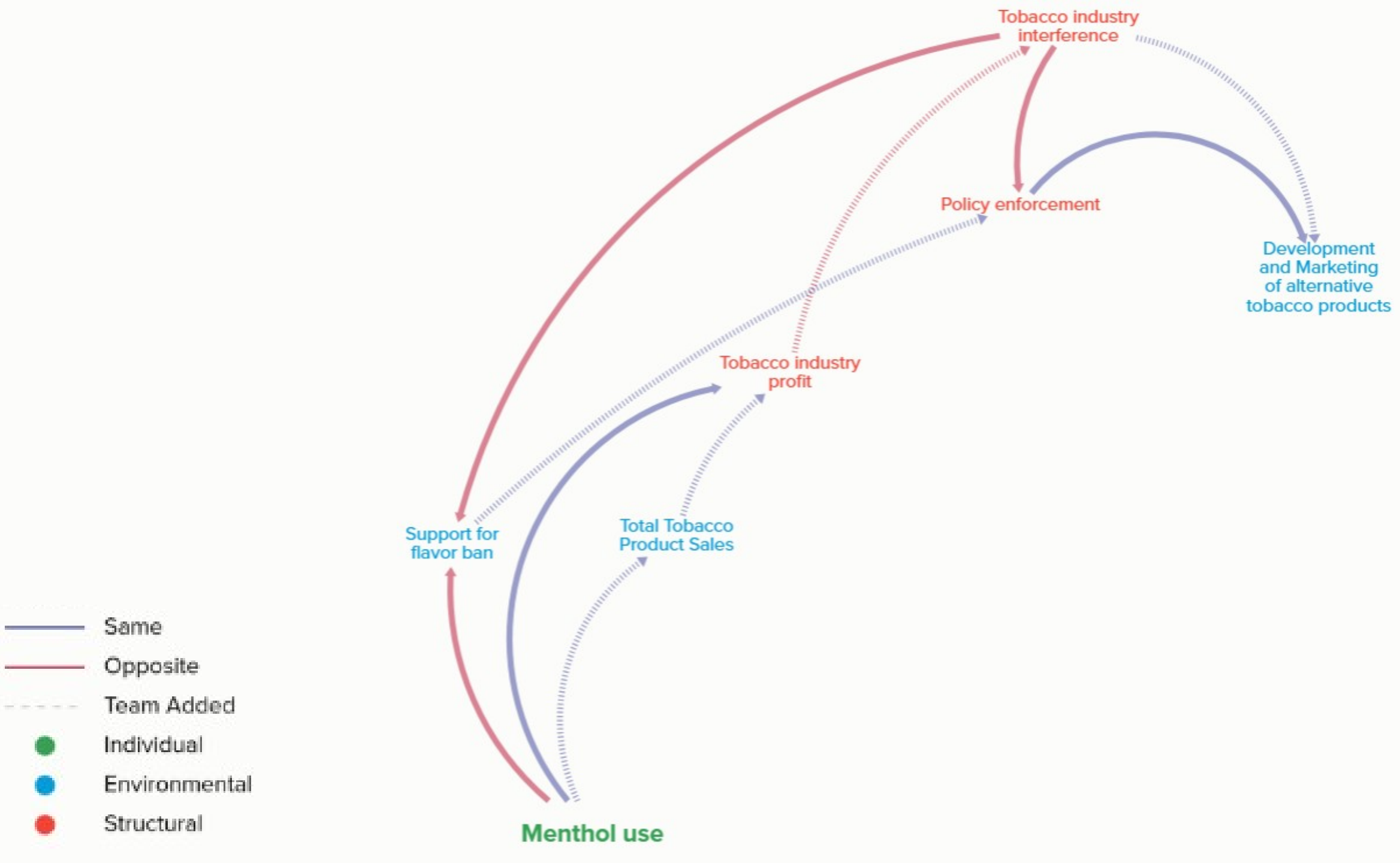
Impacts of a Ban

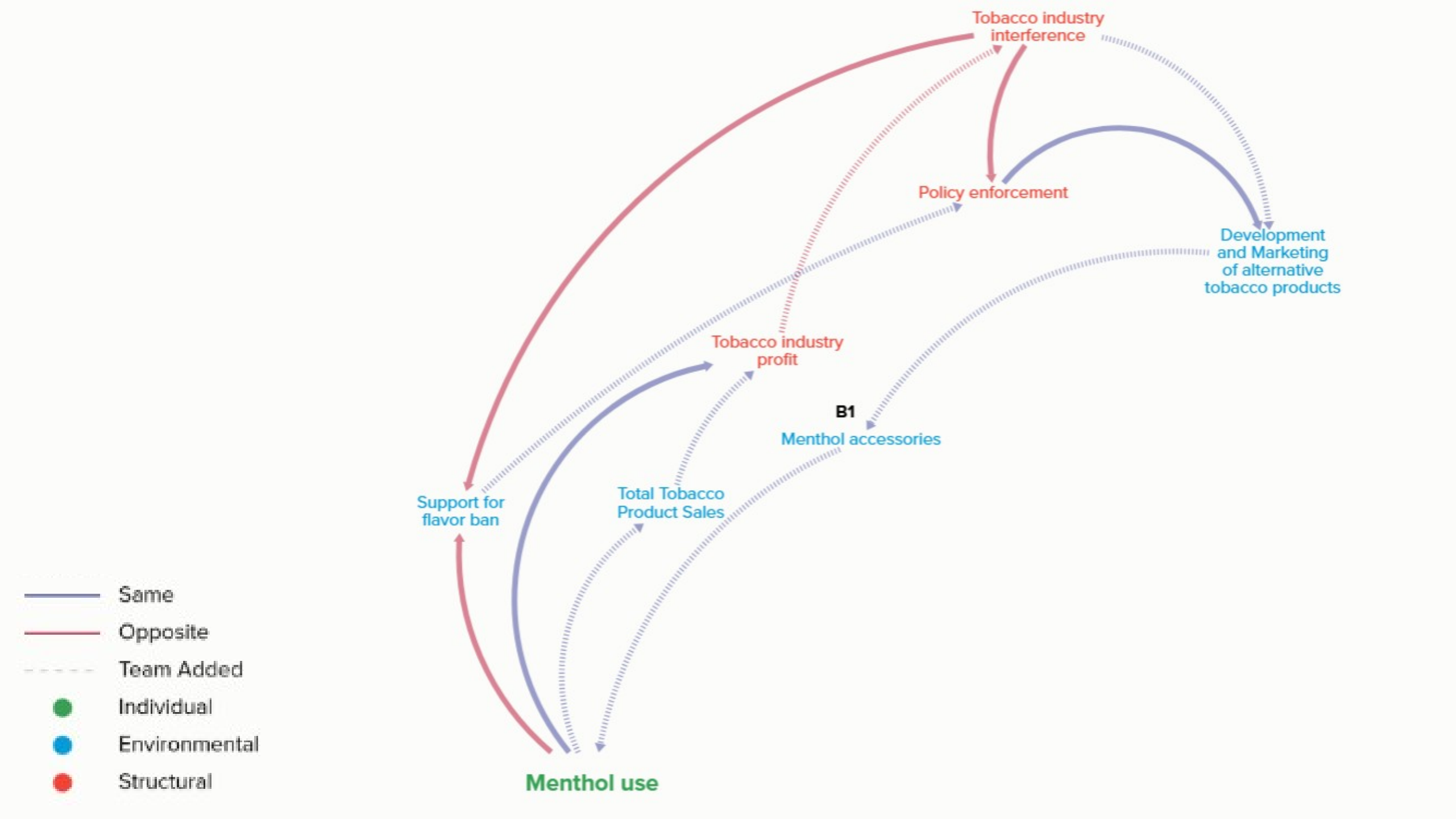


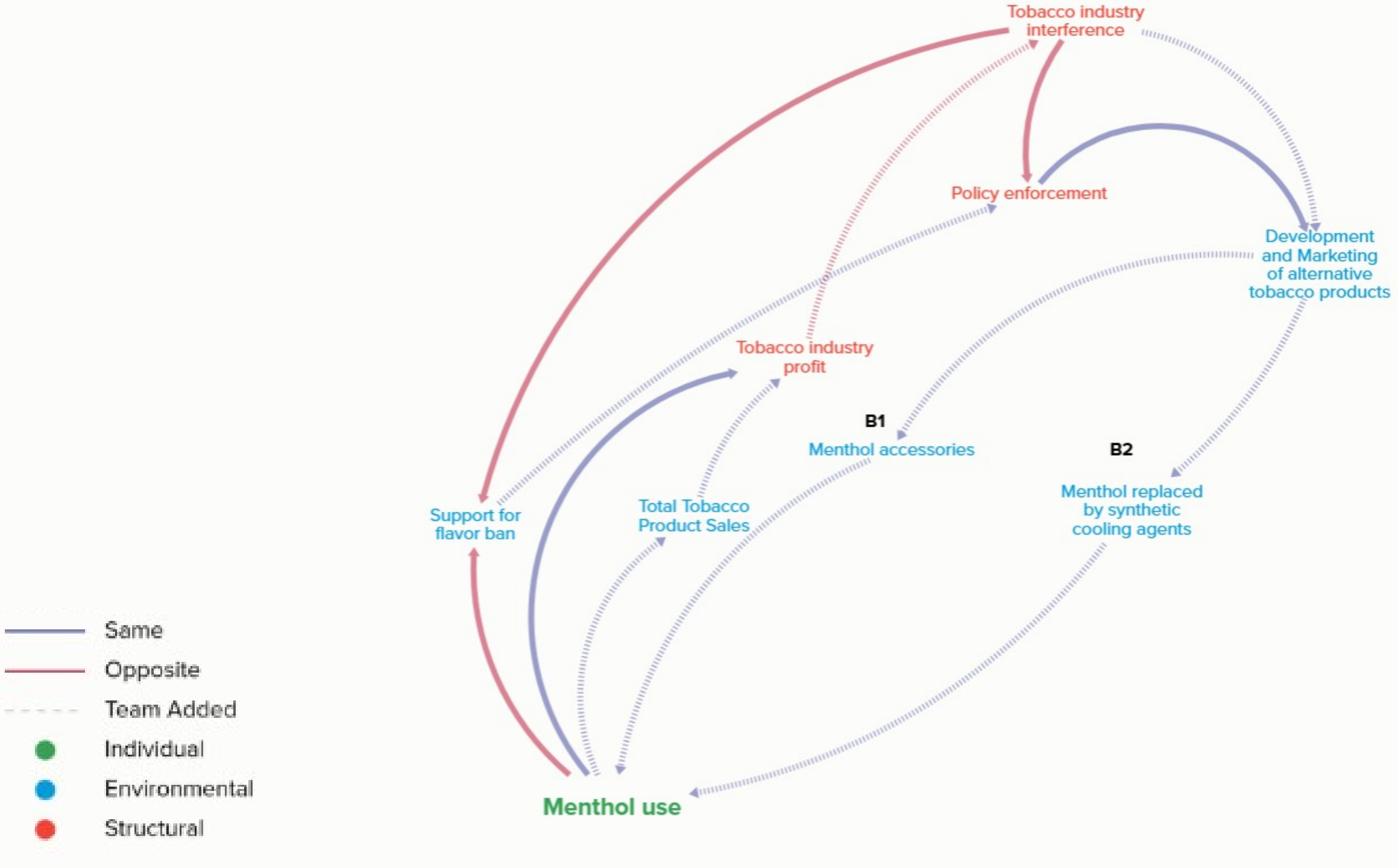
Same
Opposite
Team Added
Individual
Environmental
Structural

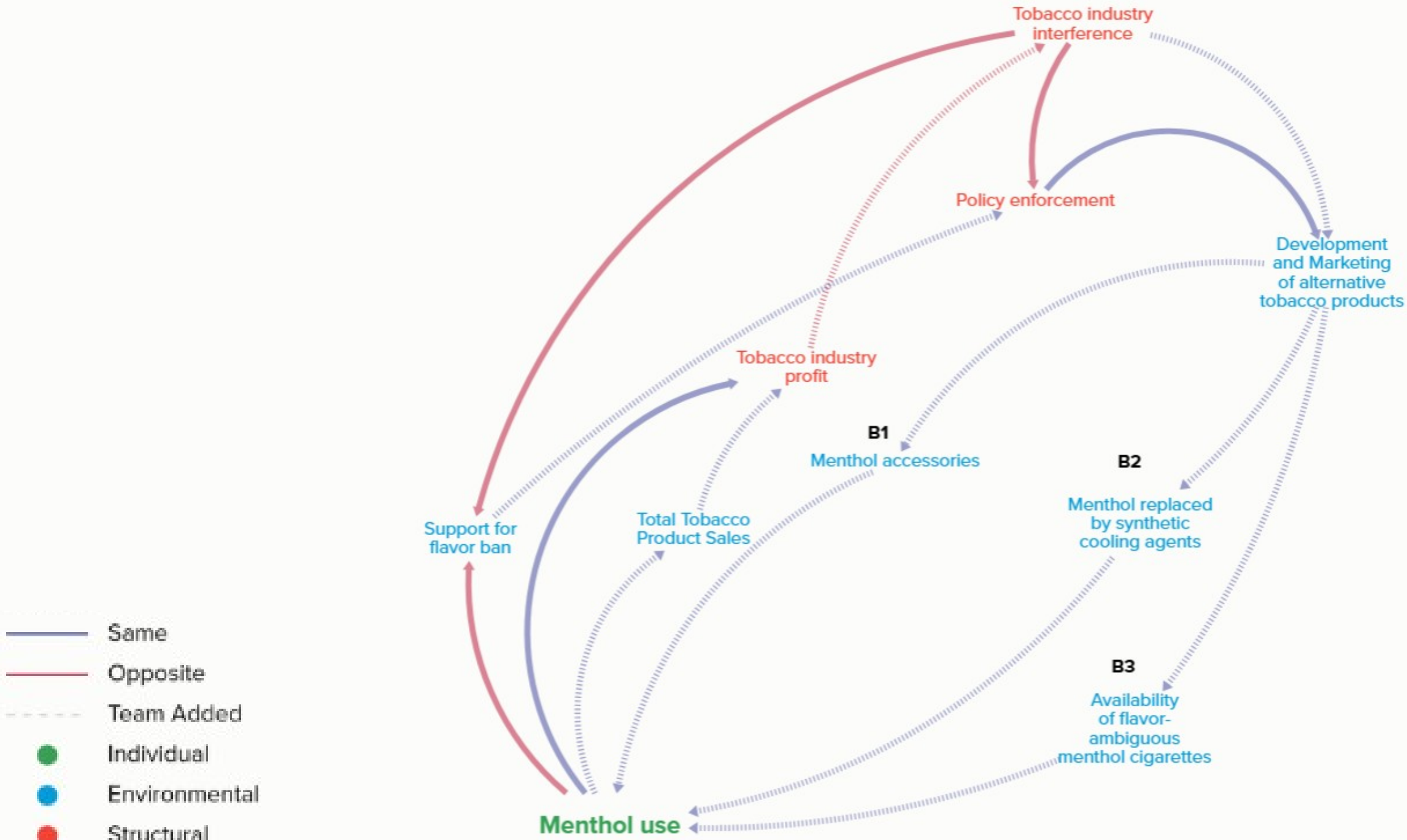

Policy enforcement
Tobacco industry profit
Support for flavor ban
Total Tobacco Product Sales
Same
Opposite
Team Added
Individual
Environmental
Structural
Menthol use

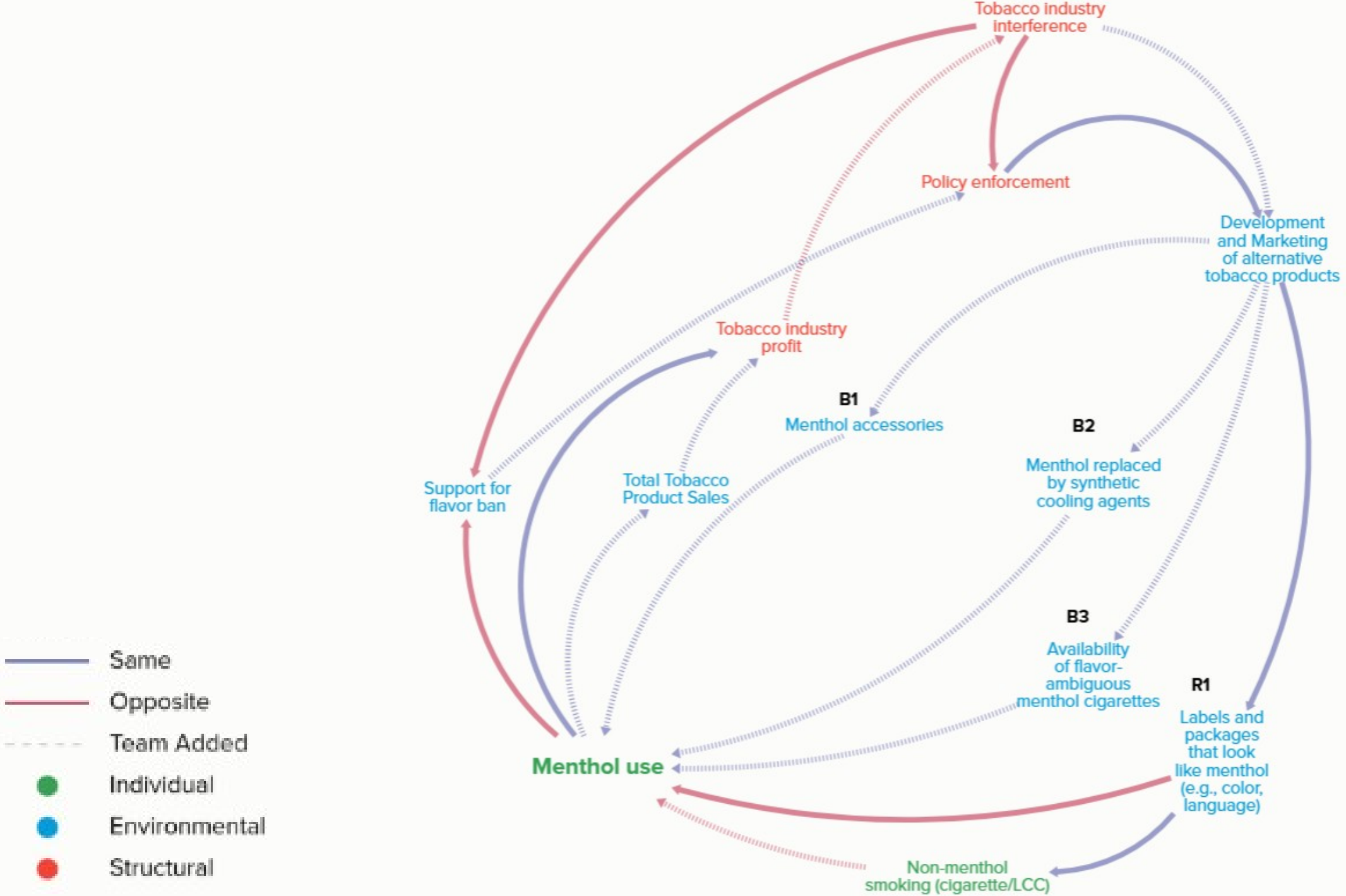

Tobacco Industry Interference
Policy enforcement
Tobacco Industry profit
Support for flavor ban
Total Tobacco Product Sales
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Tobacco industry interference
Policy enforcement
Development and Marketing of alternative tobacco products
Tobacco industry profit
Support for flavor ban
Total Tobacco Product Sales
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Tobacco industry interference
Policy enforcement
Development and Marketing of alternative tobacco products
Tobacco industry profit
B1
Menthol accessories
Support for flavor ban
Total Tobacco Product Sales
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Tobacco industry interference
Policy enforcement
Development and Marketing of alternative tobacco products
Tobacco industry profit
B1
Menthol accessories
B2
Menthol replaced by synthetic cooling agents
Support for flavor ban
Total Tobacco Product Sales
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Tobacco industry interference
Policy enforcement
Development and Marketing of alternative tobacco products
Tobacco industry profit
B1
Menthol accessories
B2
Menthol replaced by synthetic cooling agents
Support for flavor ban
Total Tobacco Product Sales
B3
Availability of flavor-ambiguous menthol cigarettes
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


Tobacco industry interference
Policy enforcement
Development and Marketing of alternative tobacco products
Tobacco industry profit
B1
Menthol accessories
B2
Menthol replaced by synthetic cooling agents
Support for flavor ban
Total Tobacco Product Sales
B3
Availability of flavor-ambiguous menthol cigarettes
R1
Labels and packages that look like menthol (e.g., color, language)
Menthol use
Non-menthol smoking (cigarette/LCC)
Same
Opposite
Team Added
Individual
Environmental
Structural

Narrative Conclusions

- Industry countermeasures are multi-level and systematic:

Narrative Conclusions

- Industry countermeasures are multi-level and systematic:
  - **Structural**: Industry interference delays enforcement

Narrative Conclusions

- Industry countermeasures are multi-level and systematic:
  - **Structural**: Industry interference delays enforcement
  - **Environmental**: Development and marketing of ban-circumventing products

Narrative Conclusions

- Industry countermeasures are multi-level and systematic:
  - **Structural**: Industry interference delays enforcement
  - **Environmental**: Development and marketing of ban-circumventing products
  - **Individual**: Targeting user perceptions and behaviors to sustain nicotine dependence and

Key Narratives



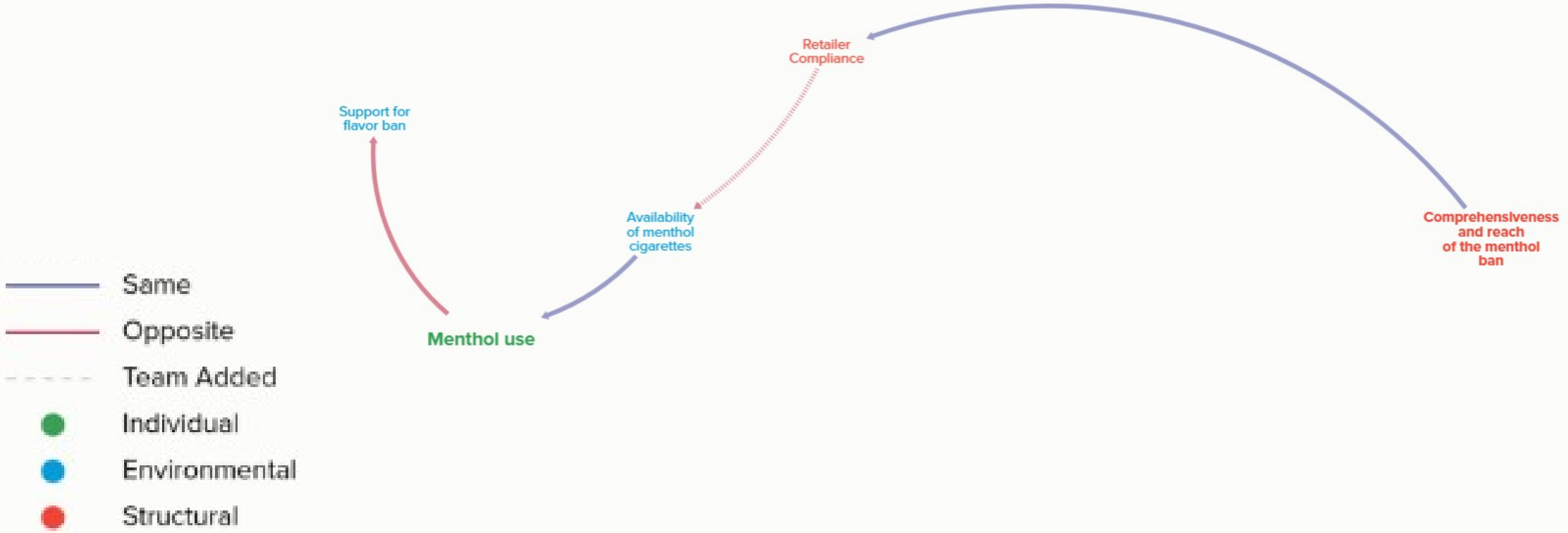
Retailer Compliance

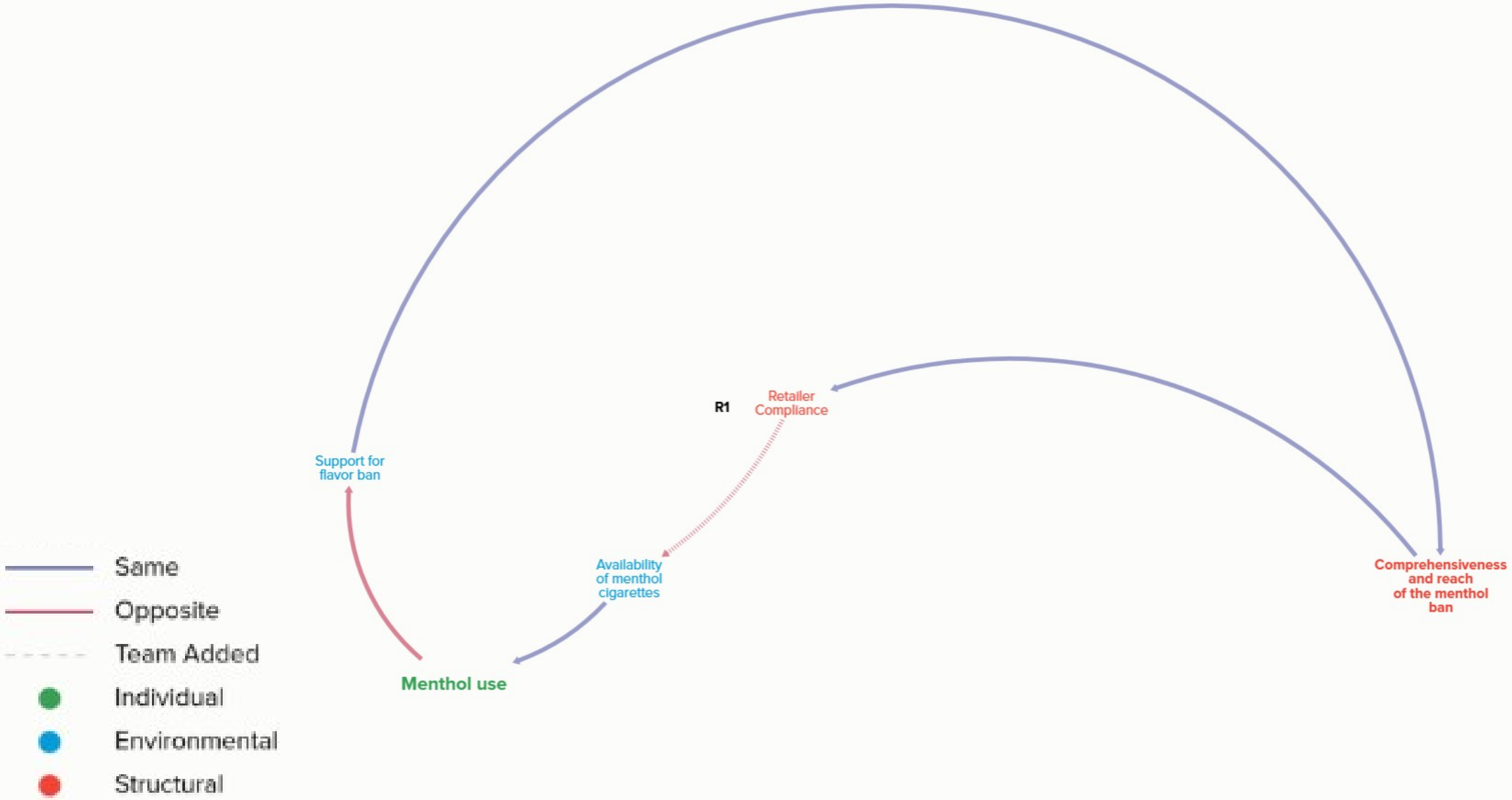


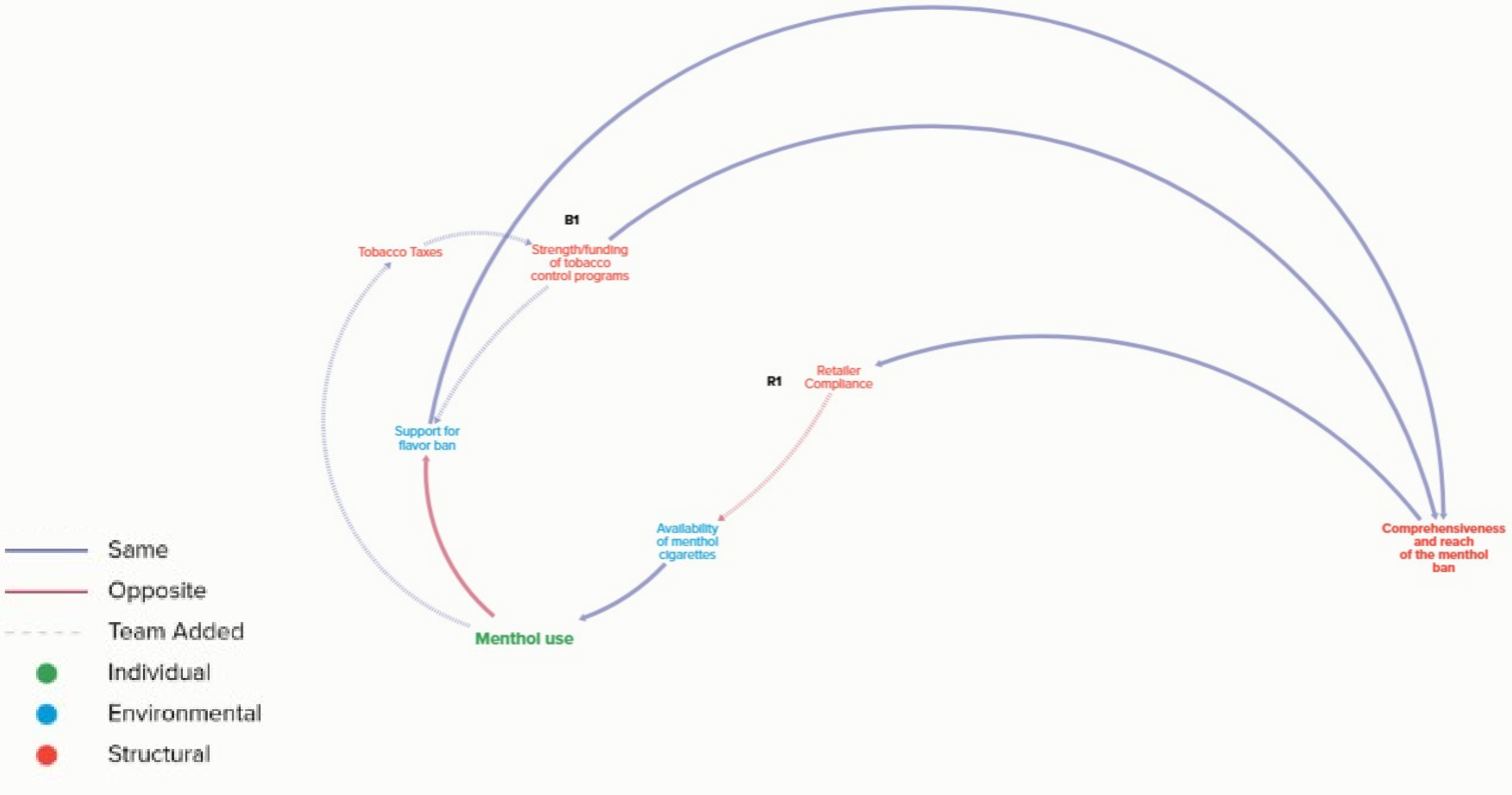
Industry Tactics

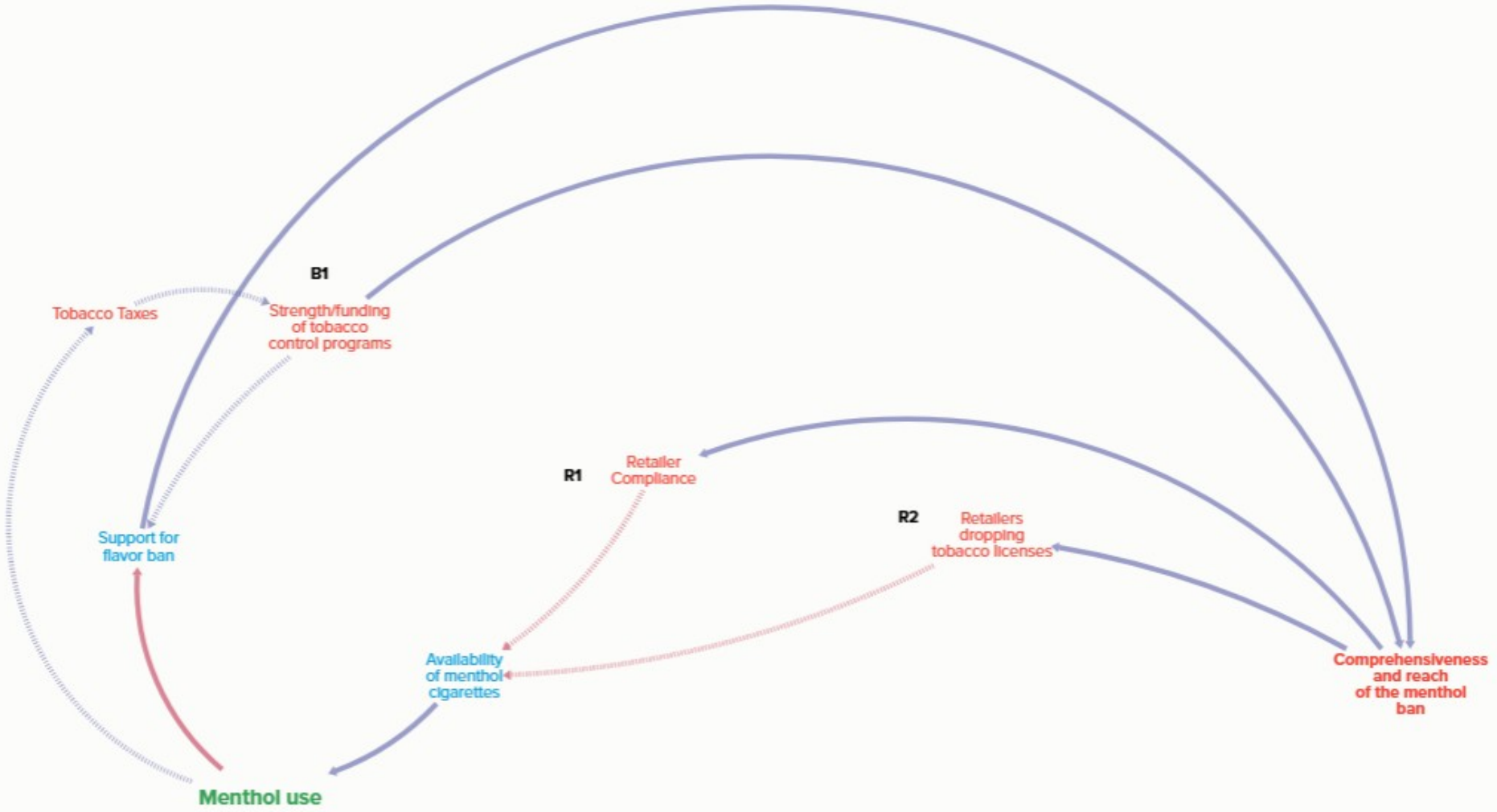


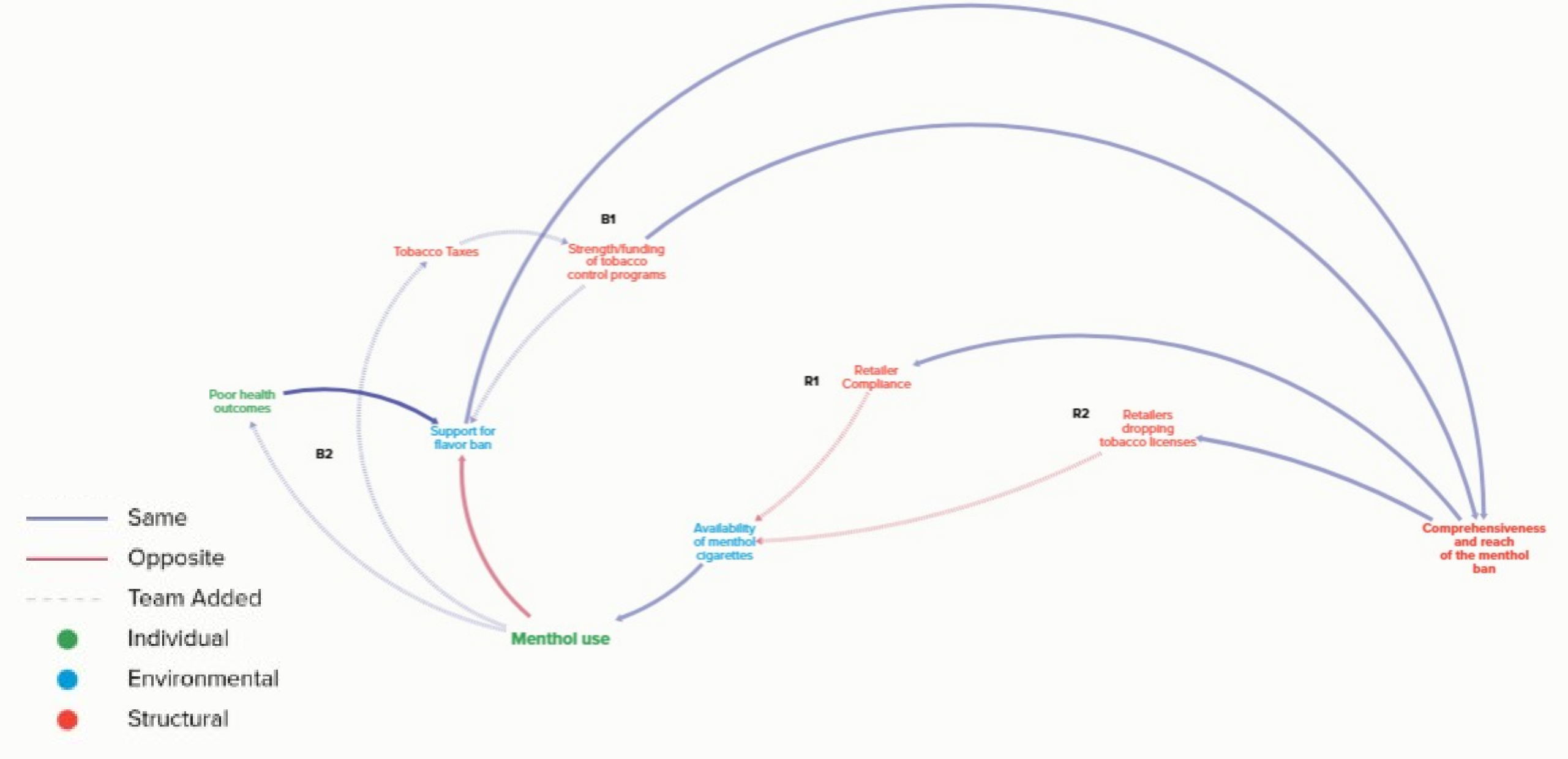
Impacts of a Ban

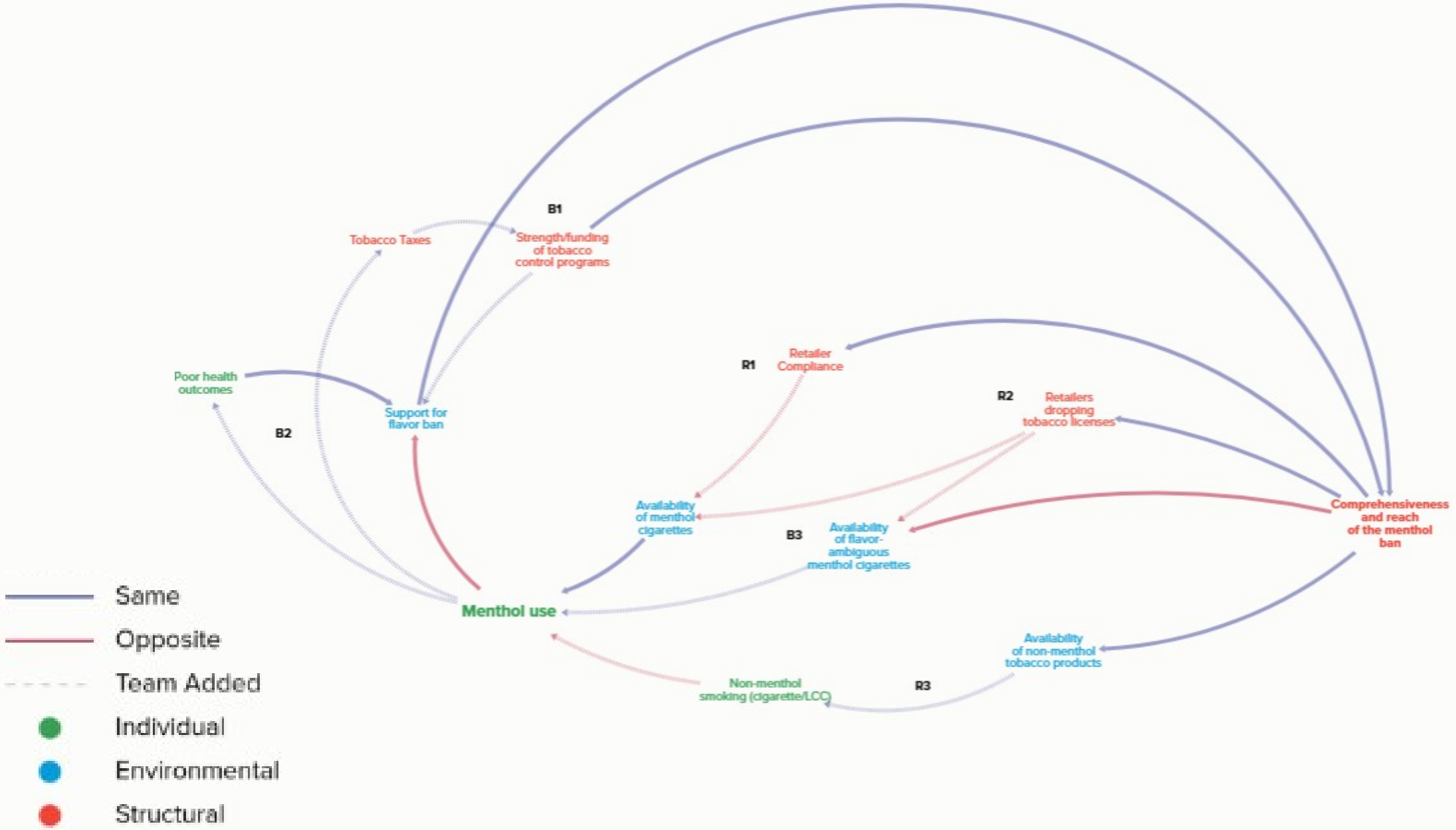

Retailer Compliance
Support for flavor ban
Availability of menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural

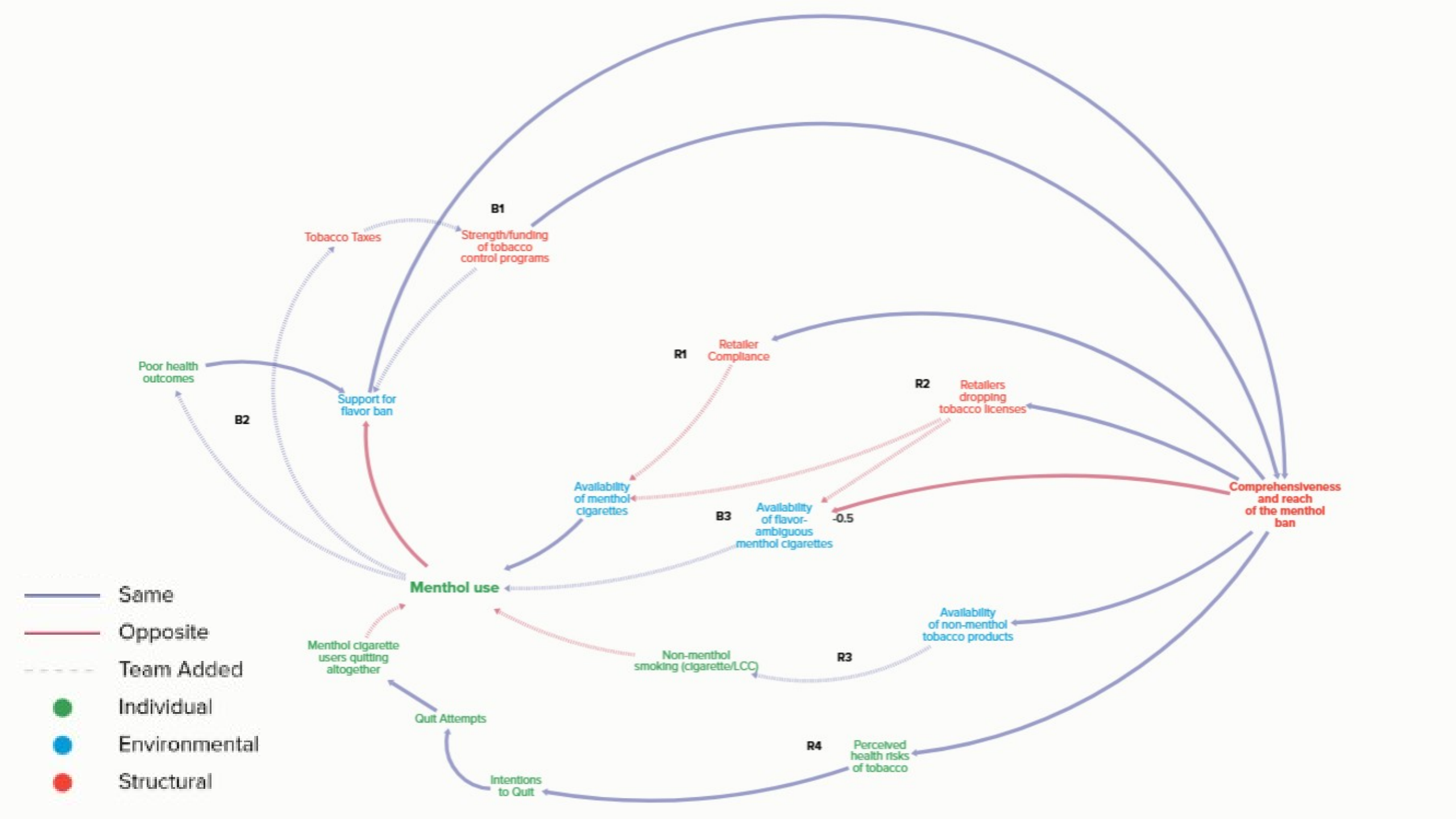

Same
Opposite
Team Added
Individual
Environmental
Structural
Support for flavor ban
R1
Retailer Compliance
Availability of menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use

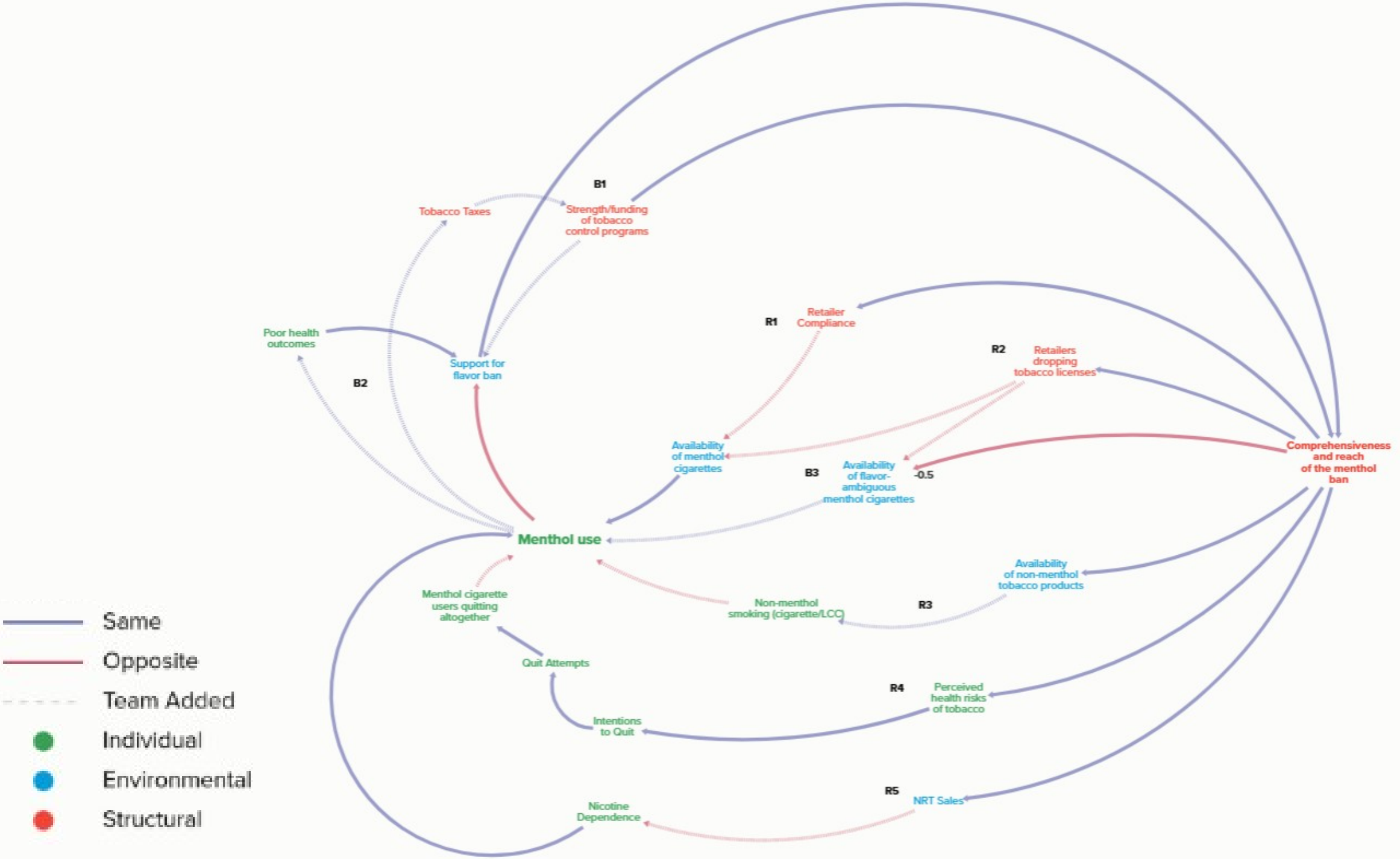

B1
Tobacco Taxes
Strength/funding of tobacco control programs
R1
Retailer Compliance
Support for flavor ban
Availability of menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


B1
Tobacco Taxes
Strength/funding of tobacco control programs
R1
Retailer Compliance
R2
Retailers dropping tobacco licenses
Support for flavor ban
Availability of menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


B1
Tobacco Taxes
Strength/funding of tobacco control programs
Poor health outcomes
Support for flavor ban
B2
R1
Retailer Compliance
R2
Retailers dropping tobacco licenses
Availability of menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use
Same
Opposite
Team Added
Individual
Environmental
Structural


B1
Tobacco Taxes
Strength/funding of tobacco control programs
Poor health outcomes
Support for flavor ban
B2
R1
Retailer Compliance
R2
Retailers dropping tobacco licenses
Availability of menthol cigarettes
B3
Availability of flavor-ambiguous menthol cigarettes
Comprehensiveness and reach of the menthol ban
Menthol use
Availability of non-menthol tobacco products
Non-menthol smoking (cigarette/LCC)
R3
Same
Opposite
Team Added
Individual
Environmental
Structural

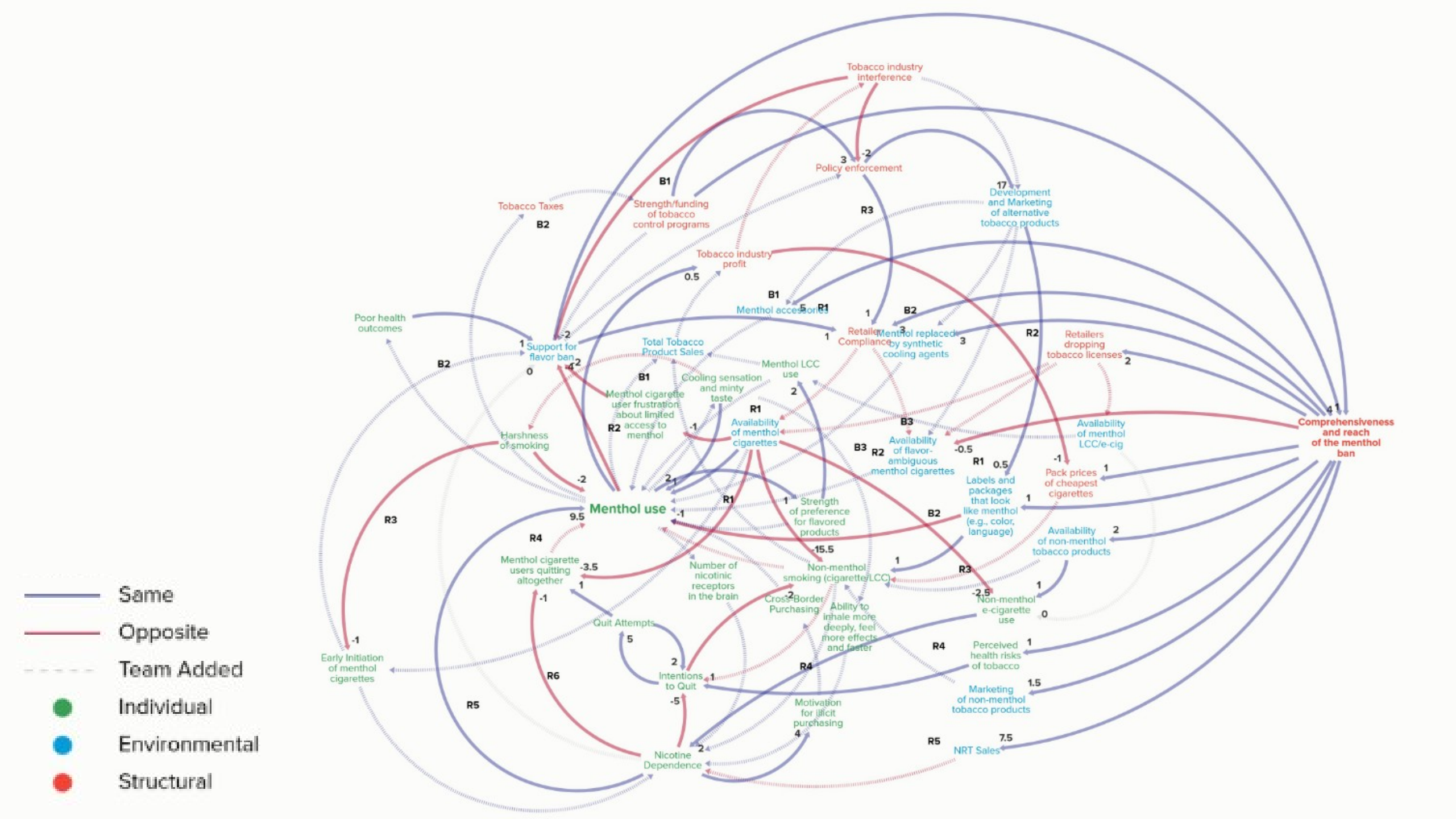

B1
Tobacco Taxes
Strength/funding of tobacco control programs
Poor health outcomes
Support for flavor ban
B2
R1
Retailer Compliance
R2
Retailers dropping tobacco licenses
Availability of menthol cigarettes
B3
Availability of flavor-ambiguous menthol cigarettes
-0.5
Comprehensiveness and reach of the menthol ban
Menthol use
Availability of non-menthol tobacco products
Menthol cigarette users quitting altogether
Non-menthol smoking (cigarette/LCC)
R3
Quit Attempts
R4
Perceived health risks of tobacco
Intentions to Quit
Same
Opposite
Team Added
Individual
Environmental
Structural


B1
Tobacco Taxes
Strength/funding of tobacco control programs
Poor health outcomes
Support for flavor ban
B2
R1
Retailer Compliance
R2
Retailers dropping tobacco licenses
Availability of menthol cigarettes
B3
Availability of flavor-ambiguous menthol cigarettes
-0.5
Comprehensiveness and reach of the menthol ban
Menthol use
Availability of non-menthol tobacco products
Menthol cigarette users quitting altogether
Non-menthol smoking (cigarette/LCC)
R3
Quit Attempts
R4
Perceived health risks of tobacco
Intentions to Quit
R5
NRT Sales
Nicotine Dependence
Same
Opposite
Team Added
Individual
Environmental
Structural

Narrative Conclusions

- Bans lead to a reduction in menthol use via multiple pathways

- Maintaining public support, risk perceptions, and cessation support are essential


Same
Opposite
Team Added
Individual
Environmental
Structural
Tobacco industry interference
Policy enforcement
Strength/funding of tobacco control programs
Tobacco Taxes
Development and Marketing of alternative tobacco products
Tobacco industry profit
Menthol accessories
Poor health outcomes
Support for flavor ban
Retailer Compliance
Menthol replaced by synthetic cooling agents
Retailers dropping tobacco licenses
Total Tobacco Product Sales
Menthol LCC use
Menthol cigarette user frustration about limited access to menthol
Cooling sensation and minty taste
Availability of menthol cigarettes
Availability of menthol LCC/e-cig
Comprehensiveness and reach of the menthol ban
Harshness of smoking
Availability of flavor-ambiguous menthol cigarettes
Pack prices of cheapest cigarettes
Labels and packages that look like menthol (e.g., color, language)
Menthol use
Strength of preference for flavored products
Availability of non-menthol tobacco products
Menthol cigarette users quitting altogether
Number of nicotinic receptors in the brain
Non-menthol smoking (cigarette/LCC)
Cross-Border Purchasing
Ability to inhale more deeply, feel more effects and faster
Non-menthol e-cigarette use
Quit Attempts
Perceived health risks of tobacco
Early Initiation of menthol cigarettes
Intentions to Quit
Marketing of non-menthol tobacco products
Motivation for illicit purchasing
Nicotine Dependence
NRT Sales
B1
B2
R1
R2
R3
R4
R5
R6

Extracted Relationships with Strong Evidence

- Menthol bans reduce menthol cigarette use

Extracted Relationships with Strong Evidence

- Menthol bans reduce menthol cigarette use
- Menthol smokers switch to non-menthol cigarettes post-ban

Extracted Relationships with Strong Evidence

- Menthol bans reduce menthol cigarette use
- Menthol smokers switch to non-menthol cigarettes post-ban
- Despite switching, total tobacco sales decrease post-ban

Extracted Relationships with Strong Evidence

- Most evidence shows no increase in illicit purchasing

Extracted Relationships with Strong Evidence

- Most evidence shows no increase in illicit purchasing
- Cross-border purchasing increases after bans

Strengths/Limitations

- Systematic coding from 141 studies reduces subjectivity
- Our depicted relationships ignore effect parameters
- Sometimes, causal links were added based on theory to bridge the gap where there was no direct evidence

Implications

- Tobacco control efforts must be multi-level and systematic to address industry tactics

Implications

- Tobacco control efforts must be multi-level and systemic to address industry tactics
  - Robust policy support

Implications

- Tobacco control efforts must be multi-level and systemic to address industry tactics
  - Robust policy support
  - Comprehensive flavored tobacco product restrictions

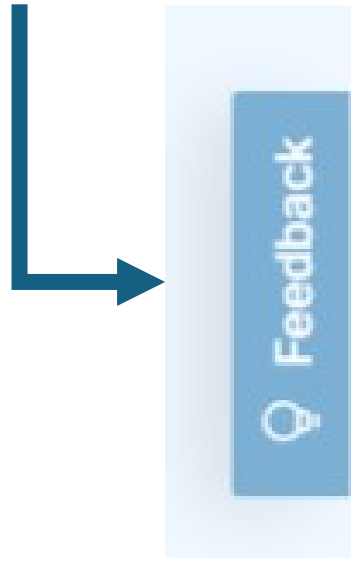
Implications

- Tobacco control efforts must be multi-level and systemic to address industry tactics
  - Robust policy support
  - Comprehensive flavored tobacco product restrictions
  - Strong enforcement and cessation support

Implications

- This work could inform simulation modeling on the impacts of menthol cigarette bans

Please provide us with feedback!

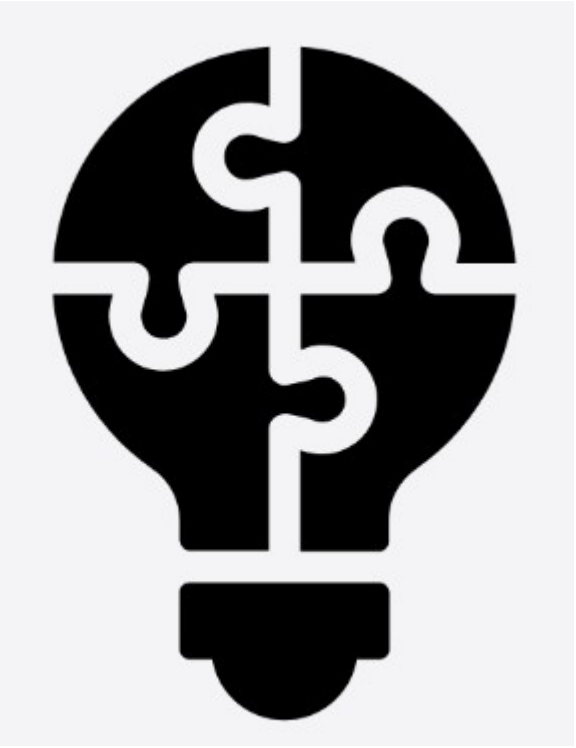


popsim.org/tools/causal-loops/menthol-cigarette-bans

Parking Lot Slides

The annotated causal loop diagrams:

- *Inform our simulation scenarios AND…*
- Integrate multidisciplinary evidence
- Develop a holistic understanding of cross-system responses to an FTP ban
- Layer in new evidence
- Illuminate knowledge gaps
- Serve as the foundation for action planning



$$Polarity\ Consensus\ Score = (S - O) - [sign(S - O) \times min(I \times 0.5, |S - O|)]$$

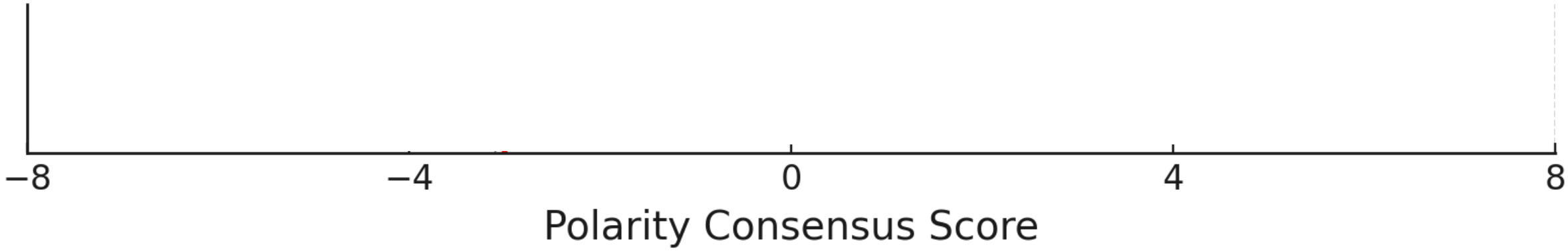
- *S* # Same relationships
- *O* # Opposite relationships
- *I* # Independent or null relationships

$$Polarity\ Consensus\ Score = (S - O) - [sign(S - O) \times min(I \times 0.5, |S - O|)]$$

- S # Same polarities
- O # Opposite polarities
- I # Independent or null polarities

Example: menthol ban (cause) and nicotine dependence (effect).

S=1, O=8, I=8

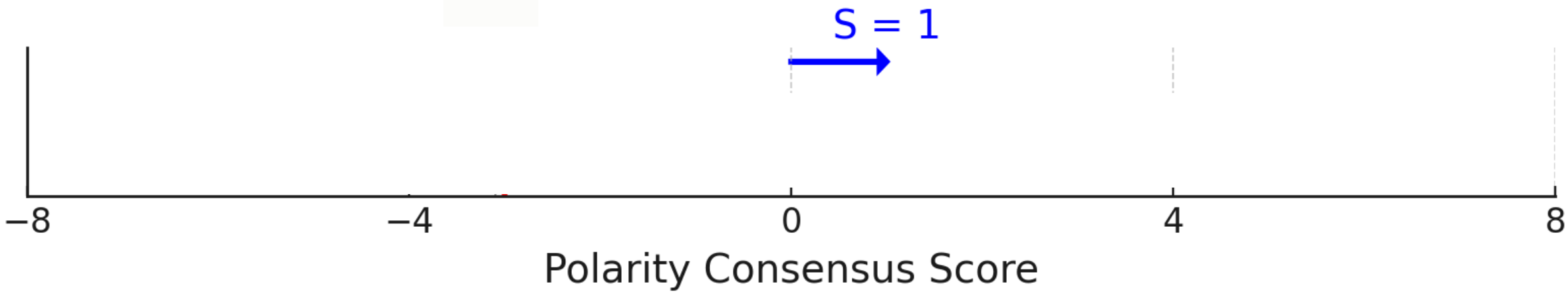


$$Polarity\ Consensus\ Score = (S - O) - [sign(S - O) \times min(I \times 0.5, |S - O|)]$$

- S # Same polarities
- O # Opposite polarities
- I # Independent or null polarities

Example: menthol ban (cause) and nicotine dependence (effect).

S=1, **O=8**, **I=8**

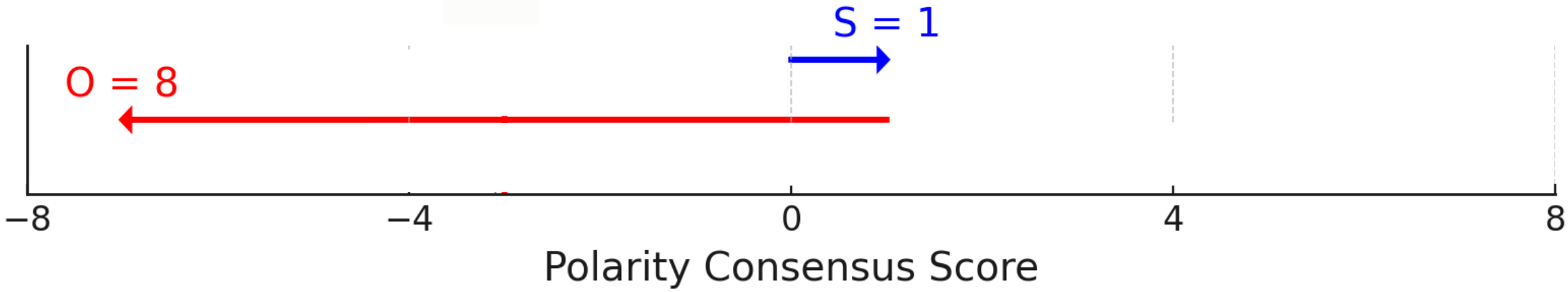


$$Polarity\ Consensus\ Score = (S - O) - [sign(S - O) \times min(I \times 0.5, |S - O|)]$$

- S # Same polarities
- O # Opposite polarities
- I # Independent or null polarities

Example: menthol ban (cause) and nicotine dependence (effect).

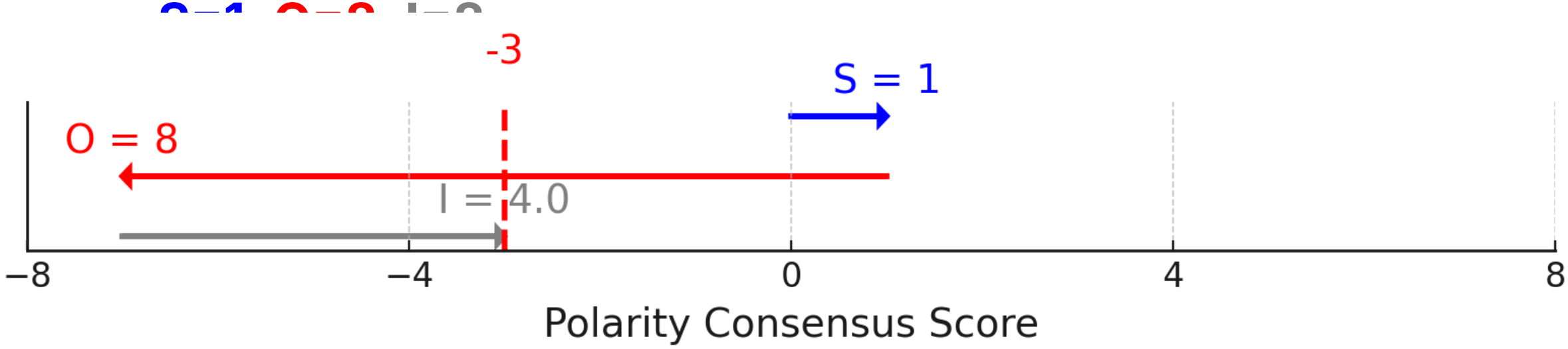
S=1, O=8, I=8



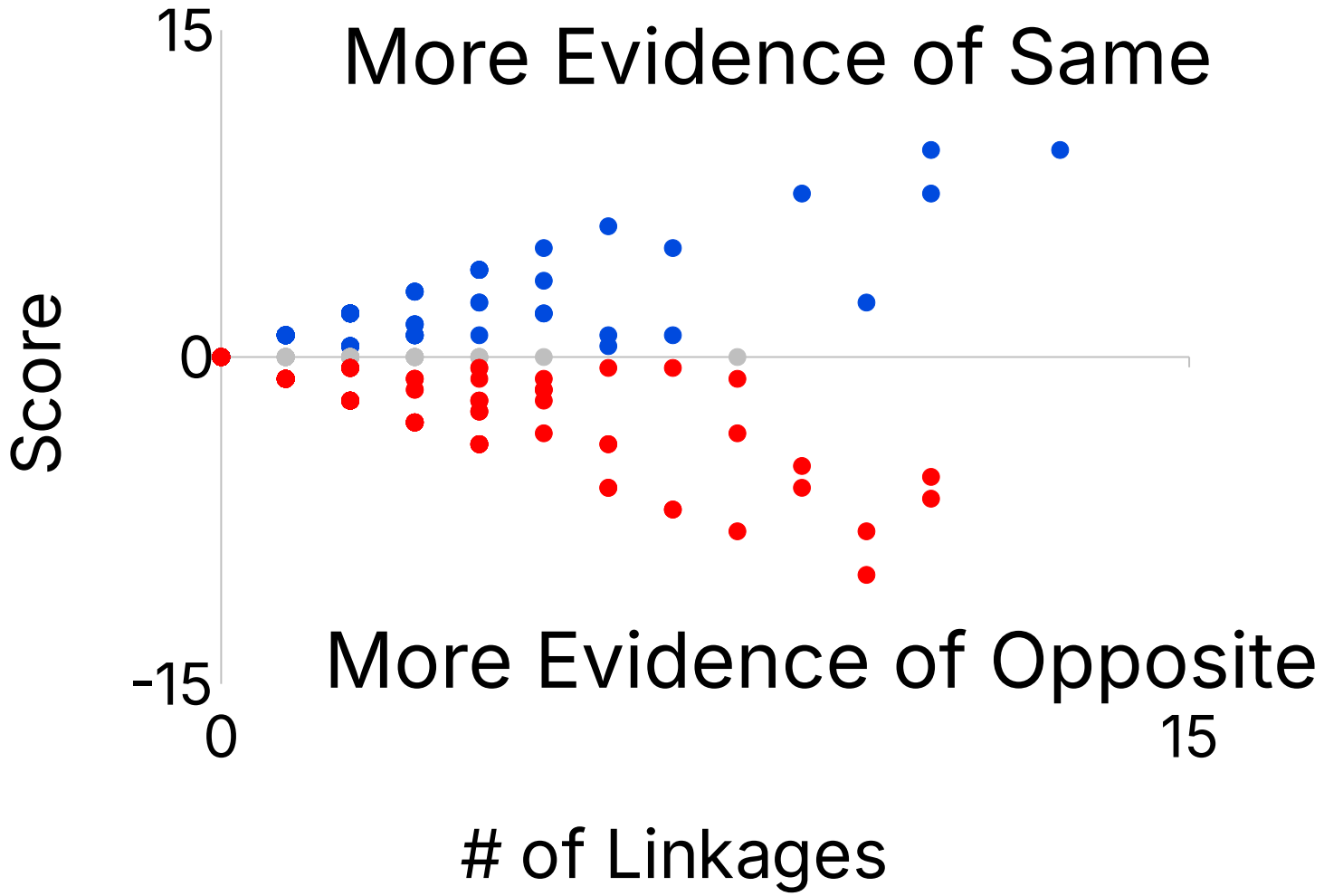
$$Polarity\ Consensus\ Score = (S - O) - [sign(S - O) \times min(I \times 0.5, |S - O|)]$$

- *S* # Same polarities
- *O* # Opposite polarities
- *I* # Independent or null polarities

Example: menthol ban (cause) and nicotine dependence (effect).



Polarity Consensus Score



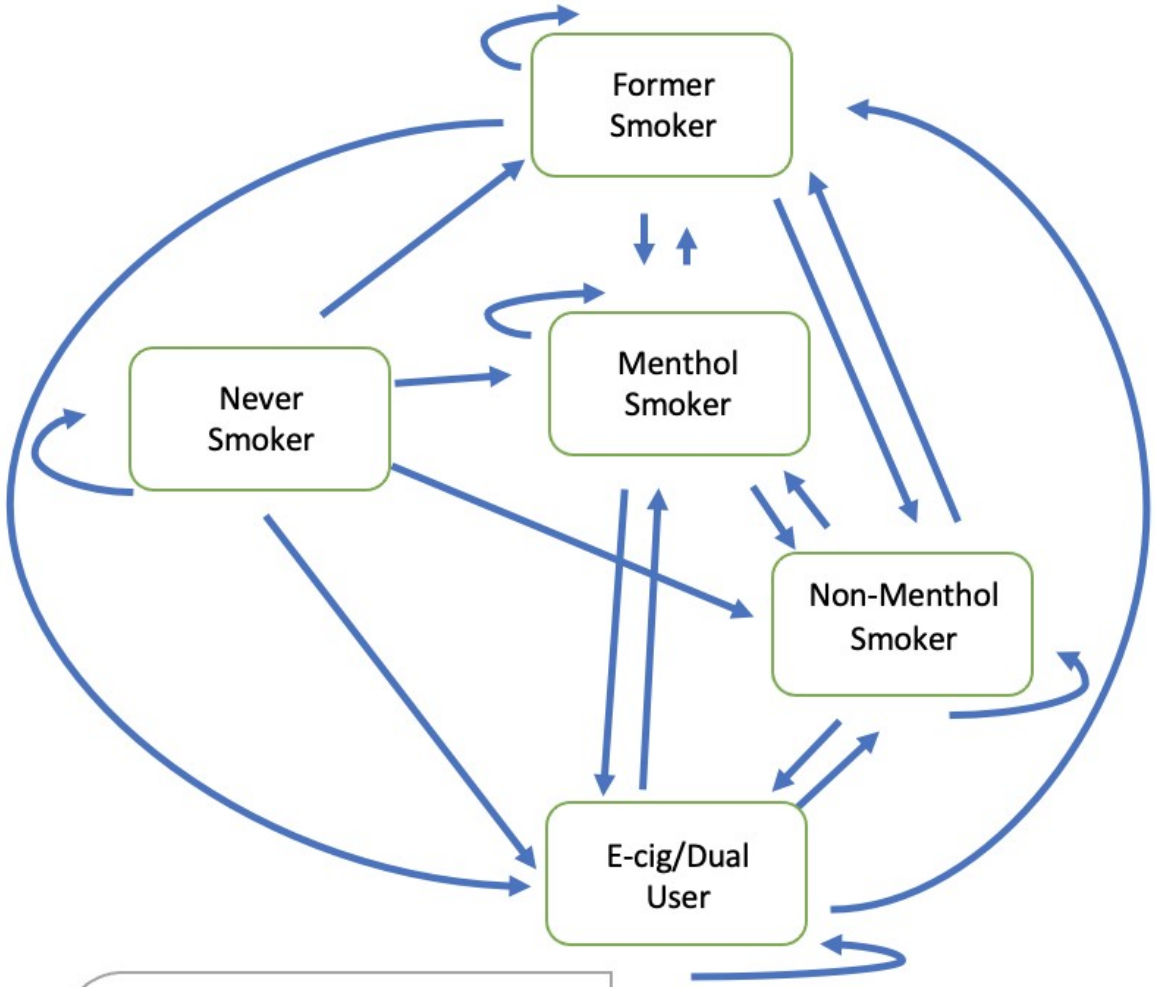
Review Process

Segment	Var1 Polarity Var2 Review
22.9% (8.0–50.3%) would reduce the amount they smoked	Menthol ban - (O) intention to reduce amount of cigarettes smoked Disagree

"Among menthol smokers (N = 62) who were asked what they would do if menthol cigarettes were banned, 31.2% (15.0–54.0%) reported they would quit smoking entirely, 22.9% (8.0–50.3%) would reduce the amount they smoked, 21.1% (9.3–41.0%) would switch to a non-menthol brand, and 16.8% (6.6–36.6%) would find a way to get menthol cigarettes." Review: Hypothetical rather than real scenario

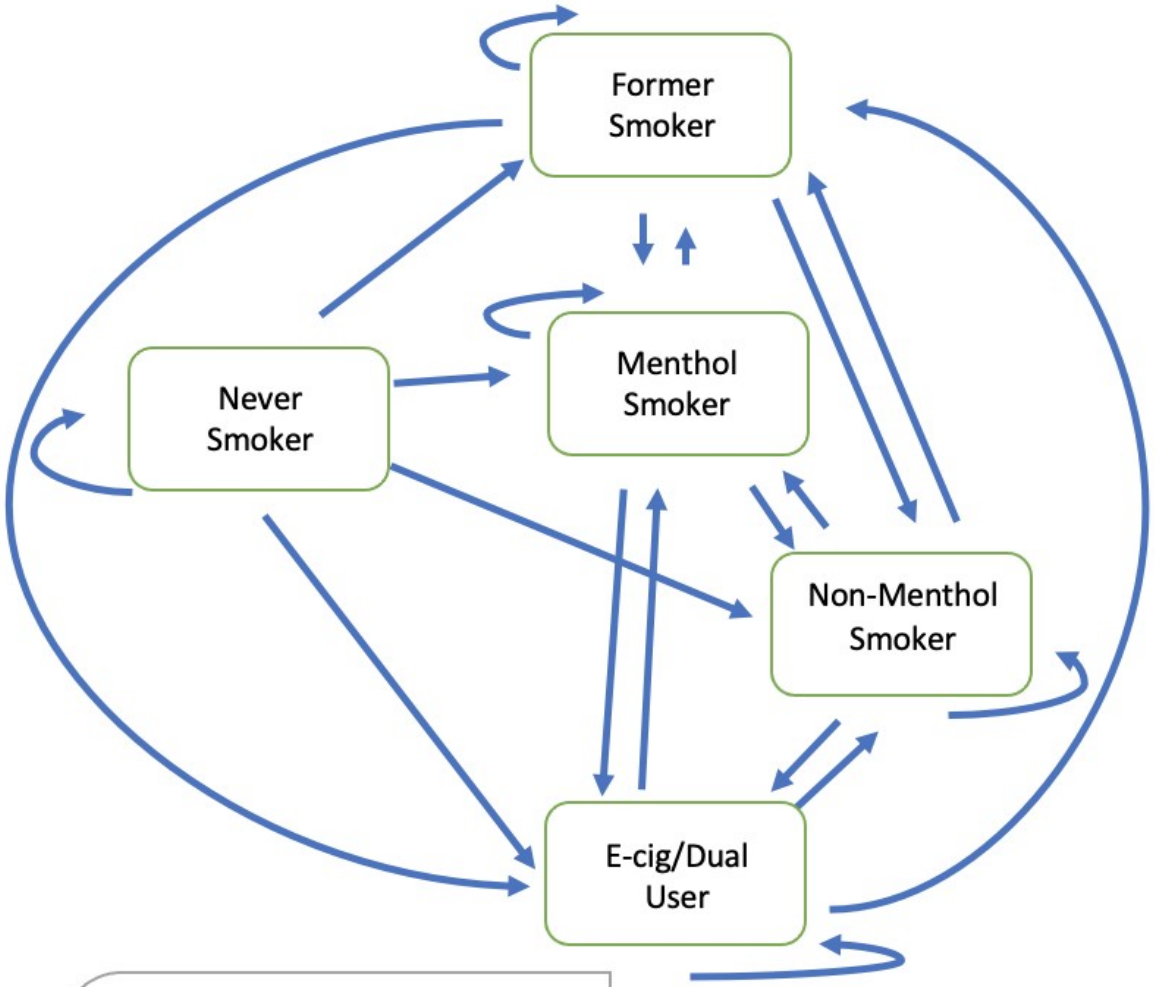
Simulation Model

- Population: PATH Study
- Second-order Markov model (PATH)
  - Initiation, cessation, and relapse rates
  - Considers individual demographics and tobacco use in current and prior year
- Focus is now on cigars (little cigars, cigarillos, traditional cigars)



Use tobacco use transition probabilities to estimate tobacco use status for each individual in each year over their lifetime

Tobacco Use States	Flavor Options
Single Product User	
LCC (with or w/o traditional cigars)	Flavored or Non-flavored
Large cigar	Flavored or Non-flavored
Cigarette	Menthol or non-menthol
ENDS	Flavored or Non-flavored
Dual Users	
LCC + Cigarette	Flavored or Non-flavored
ENDS + LCC	Flavored or Non-flavored
ENDS + Cigarette	Flavored or Non-flavored
Poly Users	
LCC, ENDS, Cigarette	Flavored or Non-flavored
Former	Flavored or Non-flavored
Never user	

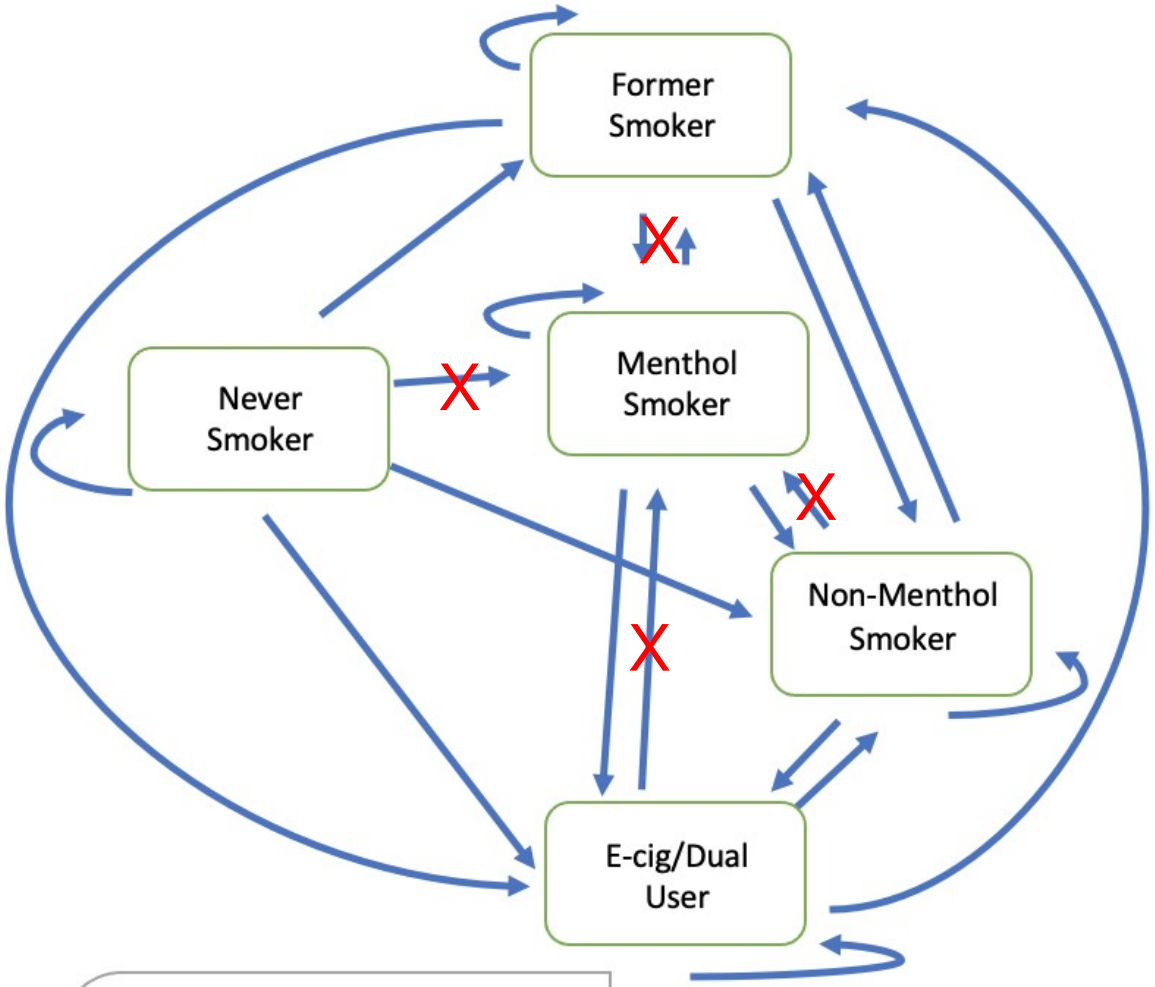


Use tobacco use transition probabilities to estimate tobacco use status for each individual in each year over their lifetime

Prevalence of tobacco use for adult population projected out 10 years

Tobacco-attributable morbidity in cohort's lifetime

Tobacco-attributable mortality in cohort's lifetime

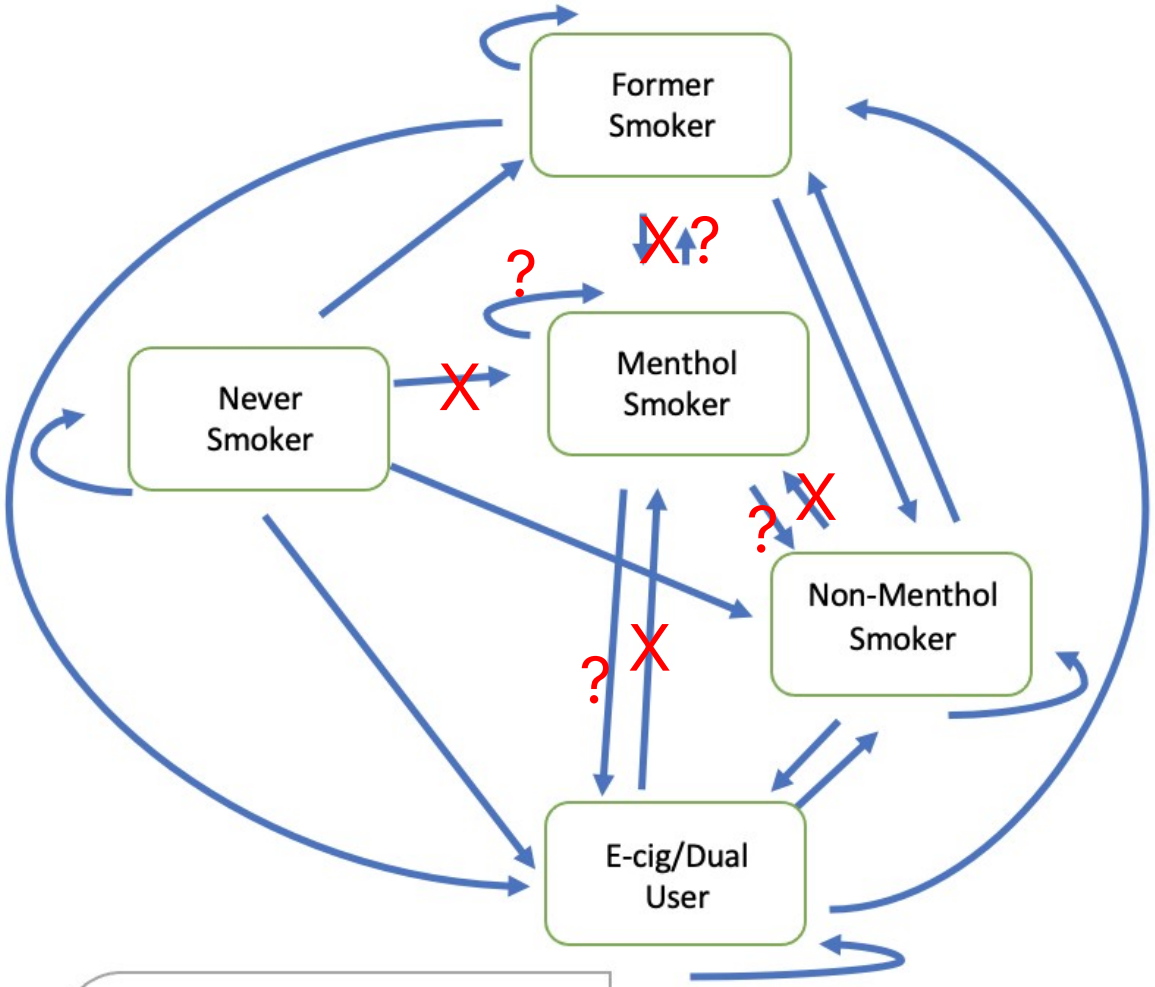


Use tobacco use transition probabilities to estimate tobacco use status for each individual in each year over their lifetime

Prevalence of tobacco use for adult population projected out 10 years

Tobacco-attributable morbidity in cohort's lifetime

Tobacco-attributable mortality in cohort's lifetime



Use tobacco use transition probabilities to estimate tobacco use status for each individual in each year over their lifetime

Prevalence of tobacco use for adult population projected out 10 years

Tobacco-attributable morbidity in cohort's lifetime

Tobacco-attributable mortality in cohort's lifetime



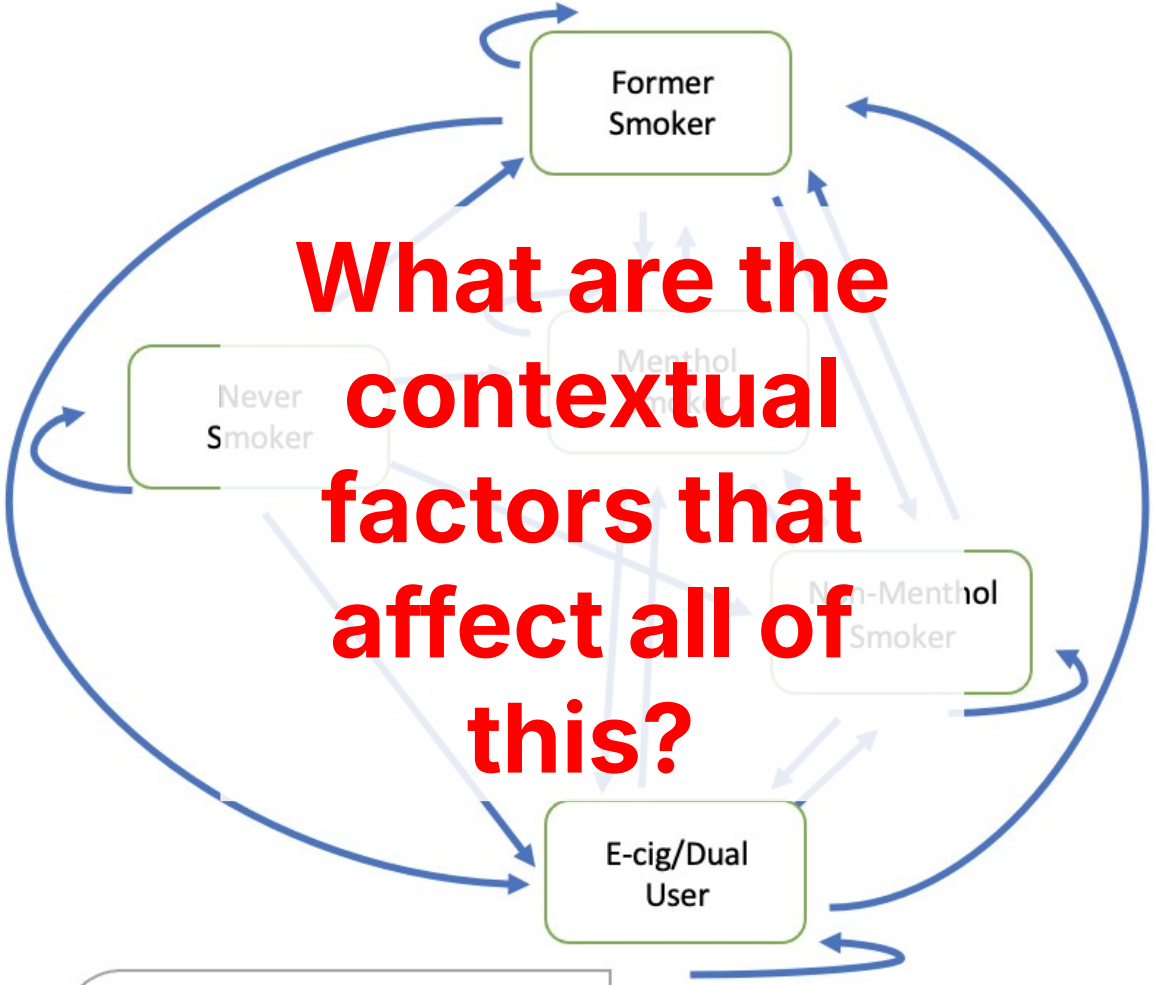
Short-term vs long-term ban effects?

In the longer-term, do people who would have taken up menthol move to other products?

Prevalence of tobacco use for adult population projected out 10 years

Tobacco-attributable morbidity in cohort's lifetime

Tobacco-attributable mortality in cohort's lifetime



What are the contextual factors that affect all of this?

Use tobacco use transition probabilities to estimate tobacco use status for each individual in each year over their lifetime

Prevalence of tobacco use for adult population projected out 10 years

Tobacco-attributable morbidity in cohort's lifetime

Tobacco-attributable mortality in cohort's lifetime

Causal Linkages Extraction

